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In [4]: # =====
# Multimodal ResNet-50 with Metadata and 7-Fold CV ≈ 70% Train / 15% Val / 15% Test
# - Concise subgroup reports (Gender, Age buckets, Skin tone)
# - Fairness metrics: WGA, EOD with min-support filtering
# - Pooled (weighted) subgroup overall accuracies across folds
# =====

import os, time, json, warnings
from copy import deepcopy
from collections import defaultdict
import numpy as np
import pandas as pd

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, Dataset
from torchvision import transforms, models
from PIL import Image

from sklearn.model_selection import StratifiedKFold, StratifiedShuffleSplit
from sklearn.metrics import (
    classification_report, accuracy_score, cohen_kappa_score,
    matthews_corrcoef, roc_auc_score
)
from sklearn.preprocessing import label_binarize
from torch.amp import GradScaler, autocast

warnings.filterwarnings('ignore')
torch.backends.cudnn.benchmark = True

# Optional FLOPs (best effort)
try:
    from thop import profile
    HAVE_THOP = True
except Exception:
    HAVE_THOP = False

# -----
# 1) CONFIG – UPDATE THESE TWO PATHS
# -----
class Config:
    DATA_DIR = r"D:\AIUB\Neural\Synthetic_Skin_Disease_Dataset" # <-- update t
    ORIGINAL_METADATA_CSV = r"D:\AIUB\Neural\complete_metadata_m.csv" # <-- update

    RESULTS_DIR = "results_Multimodal_ResNet50_KFold"
    MODEL_NAME = "ResNet-50" # Changed from DINOv3
    IMG_SIZE = 224
    BATCH_SIZE = 128
    LEARNING_RATE = 1e-4
    WEIGHT_DECAY = 2e-5
    EPOCHS = 200
    NUM_CLASSES = 5

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DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
USE_AMP = True
NUM_WORKERS = 0
PATIENCE = 20
WARMUP_EPOCHS = 10
LABEL_SMOOTHING = 0.1
PARTIALLY_UNFREEZE = False # set True to unfreeze last block of ResNet-50 (Laye

# K-Fold settings for ~70/15/15 overall
N_SPLITS = 7 # ~1/7 ≈ 14.3% Test per fold (close to 15%)
TARGET_TEST_FRAC = 0.15 # target overall (for docs)
TARGET_VAL_FRAC = 0.15 # target overall (for docs)

# Inside each fold's remainder (~85%), choose Val so overall ≈ 15%
# val_within_remainder = TARGET_VAL_FRAC / (1 - 1/N_SPLITS)
VAL_WITHIN_REMAINDER = 0.15 / (1.0 - 1.0/7.0) # ≈ 0.175

# Metadata settings
META_DROPOUT = 0.3
META_HIDDEN_DIM = 128
META_OUT_DIM = 256

# Fairness: minimum subgroup size for stable WGA/EOD & pooled tables
MIN_SUPPORT = 15

config = Config()
os.makedirs(config.RESULTS_DIR, exist_ok=True)
print("Device:", config.DEVICE)
print(f"Model: {config.MODEL_NAME}")
print(f"K-Fold: N_SPLITS={config.N_SPLITS} (~Test {100/config.N_SPLITS:.1f}%), Val_

# -----
# 2) CSV STANDARDIZATION
# -----
def standardize_metadata_columns(df: pd.DataFrame) -> pd.DataFrame:
    """
    Accepts a CSV with columns like:
        filepath, split, label, age, dominant_gender, dominant_race
    and returns standardized columns:
        filename, label, age_group, gender, skin_tone
    Keeps/deduces split if provided; K-Fold will override per fold.
    """
    df = df.copy()

    if 'filepath' in df.columns:
        df['filepath'] = df['filepath'].astype(str).str.replace('\\', '/', regex=Fa

# filename
if 'filename' not in df.columns:
    if 'filepath' in df.columns:
        df['filename'] = df['filepath'].apply(lambda p: os.path.basename(str(p)
    else:
        raise ValueError("CSV must include 'filepath' or 'filename'.")

# Label

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if 'label' not in df.columns and 'filepath' in df.columns:
    df['label'] = df['filepath'].str.split('/').str[1]
df['label'] = df['label'].astype(str)

# age_group (bucket by numeric age if present)
if 'age_group' not in df.columns:
    if 'age' in df.columns:
        age_num = pd.to_numeric(df['age'], errors='coerce')
        bins = [-1, 17, 29, 44, 64, 200] # child, young_adult, adult, middle_aged, senior
        names = ['child', 'young_adult', 'adult', 'middle_aged', 'senior']
        df['age_group'] = pd.cut(age_num, bins=bins, labels=names).astype(str).
    else:
        df['age_group'] = 'unknown'

# gender
if 'gender' not in df.columns:
    if 'dominant_gender' in df.columns:
        g = df['dominant_gender'].astype(str).str.strip().str.lower()
        df['gender'] = g.map({'man': 'male', 'male': 'male', 'woman': 'female', 'fema
    else:
        df['gender'] = 'unknown'

# skin_tone / race
if 'skin_tone' not in df.columns:
    if 'dominant_race' in df.columns:
        df['skin_tone'] = df['dominant_race'].astype(str).str.strip().str.lower
    else:
        df['skin_tone'] = 'unknown'

required = ['filename', 'label', 'age_group', 'gender', 'skin_tone']
missing = [c for c in required if c not in df.columns]
if missing:
    raise ValueError(f"Missing required columns after standardization: {missing}

return df

# -----
# 3) DATASET & VOCAB
# -----
class GPUGAugOnlyNorm(nn.Module):
    def __init__(self):
        super().__init__()
        self.normalize = transforms.Normalize(mean=[0.485, 0.456, 0.406],
                                                std=[0.229, 0.224, 0.225])

    @torch.no_grad()
    def forward(self, x): return self.normalize(x)

class MetaVocab:
    def __init__(self, df_train, cols=('age_group', 'gender', 'skin_tone')):
        self.cols = cols
        self.unk_token = "<UNK>"
        self.vocabs = {}
        for c in cols:
            uniq = sorted([self.unk_token] + sorted(df_train[c].dropna().astype(str)
            self.vocabs[c] = {v:i for i,v in enumerate(uniq)}

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        self.sizes = {c: len(self.vocabs[c]) for c in cols}
    def encode_row(self, row):
        out = {}
        for c in self.cols:
            v = str(row[c]) if pd.notnull(row[c]) else self.unk_token
            out[c] = self.vocabs[c].get(v, self.vocabs[c][self.unk_token])
        return out

class MultimodalDermDataset(Dataset):
    """
    Uses 'filepath' if present (relative to DATA_DIR or absolute),
    else DATA_DIR/split/label/filename (split provided per fold).
    """
    def __init__(self, config, df, class_to_idx, meta_vocab, transform=None):
        self.config = config
        self.df = df.reset_index(drop=True)
        self.class_to_idx = class_to_idx
        self.transform = transform
        self.meta_vocab = meta_vocab

        self.image_paths, self.labels, self.meta = [], [], []
        use_filepath = 'filepath' in self.df.columns
        for _, row in self.df.iterrows():
            if use_filepath and pd.notnull(row.get('filepath', None)):
                raw = str(row['filepath'])
                p = raw if os.path.isabs(raw) else os.path.join(config.DATA_DIR, raw)
            else:
                # Expect a 'split' column already assigned for this fold
                p = os.path.join(config.DATA_DIR, str(row['split']), str(row['label']))
            self.image_paths.append(p)
            self.labels.append(self.class_to_idx[str(row['label'])])
            self.meta.append(self.meta_vocab.encode_row(row))

        # Preload images (disable if memory constrained)
        self.images = []
        for p in self.image_paths:
            with Image.open(p).convert("RGB") as im:
                self.images.append(im.copy())

        self.class_names = sorted(class_to_idx.keys())

    def __len__(self): return len(self.labels)
    def __getitem__(self, idx):
        image = self.images[idx]
        if self.transform: image = self.transform(image)
        return image, self.labels[idx], self.meta[idx]

# -----
# 4) MODEL (ResNet-50 + Metadata Fusion) -- MODIFIED SECTION
# -----
class MetadataEncoder(nn.Module):
    def __init__(self, meta_vocab, emb_dims=None, hidden=128, out_dim=256, dropout=0.5):
        super().__init__()
        sizes = meta_vocab.sizes
        if emb_dims is None:

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        emb_dims = {
            'age_group': min(16, max(4, sizes['age_group'])),
            'gender': min(8, max(3, sizes['gender'])),
            'skin_tone': min(16, max(4, sizes['skin_tone'])),
        }
        self.emb_age = nn.Embedding(sizes['age_group'], emb_dims['age_group'])
        self.emb_gen = nn.Embedding(sizes['gender'], emb_dims['gender'])
        self.emb_tone = nn.Embedding(sizes['skin_tone'], emb_dims['skin_tone'])
        concat_dim = emb_dims['age_group'] + emb_dims['gender'] + emb_dims['skin_tone']
        self.mlp = nn.Sequential(
            nn.Linear(concat_dim, hidden), nn.GELU(), nn.Dropout(dropout),
            nn.Linear(hidden, out_dim), nn.GELU()
        )
    def forward(self, meta_idx):
        a = self.emb_age(meta_idx['age_group'])
        g = self.emb_gen(meta_idx['gender'])
        s = self.emb_tone(meta_idx['skin_tone'])
        return self.mlp(torch.cat([a, g, s], dim=-1))

class ResNet50Multimodal(nn.Module):
    """
    A multimodal model using a pretrained ResNet-50 backbone.
    Features from the image backbone are concatenated with features
    from the metadata encoder and passed to a final classifier.
    """
    def __init__(self, num_classes, meta_vocab, meta_hidden_dim, meta_out_dim, meta_dropout):
        super().__init__()
        # Image backbone (ResNet-50)
        self.backbone = models.resnet50(weights=models.ResNet50_Weights.IMAGENET1K_V1)

        # Freeze all backbone parameters initially
        for param in self.backbone.parameters():
            param.requires_grad = False

        # Unfreeze the final residual block (layer4) if configured
        if config.PARTIALLY_UNFREEZE:
            for param in self.backbone.layer4.parameters():
                param.requires_grad = True

        num_ftrs = self.backbone.fc.in_features # 2048 for ResNet-50
        self.backbone.fc = nn.Identity() # Remove original classifier

        # Metadata encoder
        self.meta_encoder = MetadataEncoder(meta_vocab,
                                            hidden=meta_hidden_dim,
                                            out_dim=meta_out_dim,
                                            dropout=meta_dropout)

        # Fusion and Classifier Head
        self.classifier = nn.Sequential(
            nn.Linear(num_ftrs + meta_out_dim, 512),
            nn.BatchNorm1d(512),
            nn.GELU(),
            nn.Dropout(0.5),
            nn.Linear(512, num_classes)
        )

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def forward(self, images, meta_idx):
    # Image stream
    self.backbone.eval() if not config.PARTIALLY_UNFREEZE else self.backbone.train()
    if config.PARTIALLY_UNFREEZE:
        img_features = self.backbone(images)
    else:
        with torch.no_grad():
            img_features = self.backbone(images)

    # Metadata stream
    meta_features = self.meta_encoder(meta_idx)

    # Fusion by concatenation
    combined_features = torch.cat((img_features, meta_features), dim=1)

    # Classification
    output = self.classifier(combined_features)
    return output

# -----
# 5) TRAIN / EVAL HELPERS
# -----
def _to_device_meta(batch_meta, device):
    if isinstance(batch_meta, dict):
        ag = torch.as_tensor(batch_meta['age_group'], dtype=torch.long, device=device)
        ge = torch.as_tensor(batch_meta['gender'], dtype=torch.long, device=device)
        st = torch.as_tensor(batch_meta['skin_tone'], dtype=torch.long, device=device)
        return {'age_group': ag, 'gender': ge, 'skin_tone': st}
    else:
        ag = torch.tensor([m['age_group'] for m in batch_meta], dtype=torch.long, device=device)
        ge = torch.tensor([m['gender'] for m in batch_meta], dtype=torch.long, device=device)
        st = torch.tensor([m['skin_tone'] for m in batch_meta], dtype=torch.long, device=device)
        return {'age_group': ag, 'gender': ge, 'skin_tone': st}

def make_dataloaders_from_df(config, df_std):
    classes = sorted(df_std[df_std['split']=='Train']['label'].astype(str).unique())
    if not classes:
        classes = sorted(df_std['label'].astype(str).unique())
    class_to_idx = {c:i for i,c in enumerate(classes)}
    config.NUM_CLASSES = len(classes)

    df_vocab = df_std[df_std['split']=='Train'] if (df_std['split']=='Train').any() else df_std
    meta_vocab = MetaVocab(df_vocab, cols=('age_group', 'gender', 'skin_tone'))

    train_tfm = transforms.Compose([
        transforms.Resize((config.IMG_SIZE, config.IMG_SIZE)),
        transforms.RandomHorizontalFlip(p=0.5),
        transforms.RandomApply([transforms.ColorJitter(0.4,0.4,0.2,0.1)], p=0.8),
        transforms.RandomRotation(degrees=15),
        transforms.ToTensor(),
    ])
    eval_tfm = transforms.Compose([
        transforms.Resize((config.IMG_SIZE, config.IMG_SIZE)),
        transforms.ToTensor(),
    ])

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))

dsets = {}
for split, tfm in [('Train', train_tfm), ('Val', eval_tfm), ('Test', eval_tfm)]:
    dfi = df_std[df_std['split'] == split]
    dsets[split] = MultimodalDermDataset(config, dfi, class_to_idx, meta_vocab,

dls = {
    'Train': DataLoader(dsets['Train'], batch_size=config.BATCH_SIZE, shuffle=True,
                        num_workers=config.NUM_WORKERS, pin_memory=True),
    'Val':   DataLoader(dsets['Val'],   batch_size=config.BATCH_SIZE, shuffle=False,
                        num_workers=config.NUM_WORKERS, pin_memory=True),
    'Test':  DataLoader(dsets['Test'],  batch_size=config.BATCH_SIZE, shuffle=False,
                        num_workers=config.NUM_WORKERS, pin_memory=True),
}
sizes = {k: len(dsets[k]) for k in dsets}
return dls, sizes, classes, meta_vocab

def train_model(model, dls, sizes, num_epochs):
    start = time.time()
    best = deepcopy(model.state_dict())
    best_val = float('inf')
    patience = 0
    hist = {'train_loss':[], 'val_loss':[], 'train_acc':[], 'val_acc':[]}

    scaler = GradScaler(enabled=config.USE_AMP)
    gpu_norm = GPUAugOnlyNorm().to(config.DEVICE)
    opt = optim.AdamW(filter(lambda p: p.requires_grad, model.parameters()),
                        lr=config.LEARNING_RATE, weight_decay=config.WEIGHT_DECAY)

    def lr_lambda(ep):
        if ep < config.WARMUP_EPOCHS: return float(ep+1)/float(config.WARMUP_EPOCHS)
        return max(0.0, 0.5 * (1.0 + np.cos(np.pi * (ep - config.WARMUP_EPOCHS) / (
    sch = optim.lr_scheduler.LambdaLR(opt, lr_lambda)
    criterion = nn.CrossEntropyLoss(label_smoothing=config.LABEL_SMOOTHING)

    print("\n" + "="*80 + f"\nSTART TRAINING (Multimodal {config.MODEL_NAME}) for {
    for ep in range(num_epochs):
        for phase in ['Train', 'Val']:
            model.train() if phase=='Train' else model.eval()
            run_loss, run_correct = 0.0, 0
            for images, labels, metas in dls[phase]:
                images = images.to(config.DEVICE)
                labels = torch.as_tensor(labels, dtype=torch.long, device=config.DE
                meta_idx = _to_device_meta(metas, config.DEVICE)
                images = gpu_norm(images)
                opt.zero_grad(set_to_none=True)
                with autocast(device_type=config.DEVICE.split(':')[0], dtype=torch.
                    logits = model(images=images, meta_idx=meta_idx)
                    loss = criterion(logits, labels)
                if phase=='Train':
                    scaler.scale(loss).backward()
                    scaler.step(opt)
                    scaler.update()
                preds = logits.argmax(1)
                run_loss += loss.item() * images.size(0)

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        run_correct += (preds == labels).sum().item()
    e_loss = run_loss / max(1, sizes[phase])
    e_acc = run_correct / max(1, sizes[phase])
    if phase=='Train':
        hist['train_loss'].append(e_loss); hist['train_acc'].append(e_acc)
    else:
        hist['val_loss'].append(e_loss); hist['val_acc'].append(e_acc)
        if e_loss < best_val:
            best_val = e_loss; best = deepcopy(model.state_dict()); patience
            torch.save(model.state_dict(), os.path.join(config.RESULTS_DIR,
            else:
                patience += 1
    sch.step()
    print(f"Epoch {ep+1}/{num_epochs} | Train {hist['train_loss'][-1]:.4f}/{his
        f"Val {hist['val_loss'][-1]:.4f}/{hist['val_acc'][-1]:.4f} | LR {sch.
    if patience >= config.PATIENCE:
        print("Early stopping.")
        break
    model.load_state_dict(best)
    print("Training complete in {:.1f} minutes. Best Val Loss {:.4f}".format((time.
    return model, hist

# -----
# 6) EVALUATION (Concise reports + WGA/EOD with min-support + Pooled subgroup accur
# -----
def _print_section(title):
    print("\n" + "="*80)
    print(title.upper())
    print("="*80 + "\n")

def _print_group_report(group_title, y_true, y_pred, class_names):
    labels_full = list(range(len(class_names)))
    print(f"--- Results for Group: {group_title} ---")
    print(classification_report(y_true, y_pred,
                                labels=labels_full,
                                target_names=class_names,
                                zero_division=0))

    print()

def _group_fairness_metrics(y_true, y_pred, groups, num_classes, min_support=None):
    uniq_groups = sorted(pd.Series(groups).astype(str).unique())
    per_group_acc, per_group_tprs, counts = {}, {}, {}
    y_true = np.asarray(y_true); y_pred = np.asarray(y_pred)

    for g in uniq_groups:
        mask = (np.asarray(groups) == g)
        counts[g] = int(mask.sum())
        yt = y_true[mask]; yp = y_pred[mask]
        acc_g = (yt == yp).mean() if len(yt) > 0 else np.nan
        tprs = []
        for c in range(num_classes):
            tp = ((yt == c) & (yp == c)).sum()
            pos = (yt == c).sum()
            tprs.append(tp/pos if pos > 0 else np.nan)
        per_group_acc[g] = acc_g

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        per_group_tprs[g] = np.array(tprs, dtype=float)

    if min_support is not None:
        filt_keys = [g for g in uniq_groups if counts[g] >= min_support]
    else:
        filt_keys = uniq_groups

    if filt_keys:
        wga = float(np.nanmin([per_group_acc[g] for g in filt_keys]))
    else:
        wga = float('nan')

def macro_eod(keys, per_group_tprs, K):
    max_gap, worst = 0.0, (None, None)
    for i in range(len(keys)):
        for j in range(i+1, len(keys)):
            gi, gj = keys[i], keys[j]
            a, b = per_group_tprs[gi], per_group_tprs[gj]
            mask = ~(np.isnan(a) | np.isnan(b))
            if mask.sum() == 0:
                continue
            gap = float(np.nanmean(np.abs(a[mask] - b[mask])))
            if gap > max_gap:
                max_gap, worst = gap, (gi, gj)
    return max_gap, worst

eod_val, worst_pair = macro_eod(filt_keys, per_group_tprs, num_classes)

return {
    'per_group_acc': per_group_acc,
    'per_group_tprs': per_group_tprs,
    'counts': counts,
    'keys_used': filt_keys,
    'wga': wga,
    'eod': eod_val,
    'worst_pair': worst_pair
}

def evaluate_and_visualize(model, dl_test, class_names, df_fold):
    model.eval()
    gpu_norm = GPUAugOnlyNorm().to(config.DEVICE)
    y_true, y_pred, y_prob = [], [], []

    with torch.no_grad():
        for images, labels, metas in dl_test:
            images = images.to(config.DEVICE)
            labels = torch.as_tensor(labels, dtype=torch.long, device=config.DEVICE)
            meta_idx = _to_device_meta(metas, config.DEVICE)
            images = gpu_norm(images)
            logits = model(images=images, meta_idx=meta_idx)
            probs = torch.softmax(logits, dim=1)
            y_true += labels.cpu().numpy().tolist()
            y_pred += logits.argmax(1).cpu().numpy().tolist()
            y_prob += probs.cpu().numpy().tolist()

    y_true = np.array(y_true); y_pred = np.array(y_pred); y_prob = np.array(y_prob)

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# OVERALL (concise)
_print_section("OVERALL PERFORMANCE (ON COMBINED DATASET)")
print(classification_report(y_true, y_pred,
                            labels=list(range(len(class_names))),
                            target_names=class_names,
                            zero_division=0))

# Align Test metadata rows
df_test = df_fold[df_fold['split'] == 'Test'][['age_group', 'gender', 'skin_tone']]
if len(df_test) != len(y_true):
    m = min(len(df_test), len(y_true))
    df_test = df_test.iloc[:m].copy()
    y_true = y_true[:m]; y_pred = y_pred[:m]; y_prob = y_prob[:m]

# BY GENDER
_print_section("PERFORMANCE ANALYSIS BY: DOMINANT_GENDER")
for g_key, g_title in [('male', 'Man'), ('female', 'Woman')]:
    mask = (df_test['gender'].astype(str) == g_key).values
    if mask.sum() == 0: continue
    _print_group_report(g_title, y_true[mask], y_pred[mask], class_names)

# BY AGE BUCKETS
_print_section("PERFORMANCE ANALYSIS BY: AGE_GROUP")
def bucket(x):
    x = str(x)
    if x == 'child': return '0-17'
    if x in ('young_adult', 'adult', 'middle_aged'): return '18-59'
    if x == 'senior': return '60+'
    return 'unknown'
age_disp = df_test['age_group'].apply(bucket)
for group in ['0-17', '18-59', '60+']:
    mask = (age_disp == group).values
    if mask.sum() == 0: continue
    _print_group_report(group, y_true[mask], y_pred[mask], class_names)

# BY SKIN TONE
_print_section("PERFORMANCE ANALYSIS BY: DOMINANT_RACE")
for tone in sorted(df_test['skin_tone'].astype(str).unique()):
    if tone == 'unknown': continue
    mask = (df_test['skin_tone'].astype(str) == tone).values
    if mask.sum() == 0: continue
    _print_group_report(tone, y_true[mask], y_pred[mask], class_names)

# ----- FAIRNESS SUMMARY -----
groups_composite = (age_disp.astype(str) + "|" +
                    df_test['gender'].astype(str) + "|" +
                    df_test['skin_tone'].astype(str)).values

fair_all = _group_fairness_metrics(
    y_true, y_pred, groups_composite, num_classes=len(class_names), min_support=
)
fair_filt = _group_fairness_metrics(
    y_true, y_pred, groups_composite, num_classes=len(class_names), min_support=
)

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print("\n" + "="*80)
print(f"FAIRNESS SUMMARY (ALL GROUPS) | WGA: {fair_all['wga']:.4f} EOD (max
if fair_all['worst_pair'][0] is not None:
    print(f"Worst pair: {fair_all['worst_pair'][0]} vs {fair_all['worst_pair']
print("="*80)

print("\n" + "="*80)
print(f"FAIRNESS SUMMARY (FILTERED, n >= {config.MIN_SUPPORT}) | "
      f"WGA: {fair_filt['wga']:.4f} EOD (max TPR gap): {fair_filt['eod']:.4f}
if fair_filt['worst_pair'][0] is not None:
    print(f"Worst pair (among filtered): {fair_filt['worst_pair'][0]} vs {fai
print("Groups used (filtered):", fair_filt['keys_used'])
print("="*80)

# Scalar metrics
acc = accuracy_score(y_true, y_pred)
kappa = cohen_kappa_score(y_true, y_pred)
mcc = matthews_corrcoef(y_true, y_pred)
try:
    y_true_bin = label_binarize(y_true, classes=list(range(len(class_names))))
    roc_auc_macro = roc_auc_score(y_true_bin, y_prob, average='macro', multi_cl
except Exception:
    roc_auc_macro = float('nan')

# ----- NEW: per-subgroup accuracy & counts for pooling across folds -----
def acc_counts_from_series(series_vals):
    """returns dict: group -> (correct, total)"""
    d = {}
    s = series_vals.astype(str).values
    for g in np.unique(s):
        mask = (s == g)
        tot = int(mask.sum())
        if tot == 0: continue
        cor = int((y_true[mask] == y_pred[mask]).sum())
        d[str(g)] = (cor, tot)
    return d

pooled_info = {
    'gender': acc_counts_from_series(df_test['gender']),
    'age': acc_counts_from_series(age_disp),
    'skin_tone': acc_counts_from_series(df_test['skin_tone']),
    'composite': acc_counts_from_series(pd.Series(groups_composite)),
}

return {
    'accuracy': acc, 'roc_auc_macro': roc_auc_macro, 'kappa': kappa, 'mcc': mcc
    'wga_all': fair_all['wga'], 'eod_all': fair_all['eod'],
    'wga_filtered': fair_filt['wga'], 'eod_filtered': fair_filt['eod'],
    'pooled_groups': pooled_info # NEW: for cross-fold pooled subgroup accurac
}

# -----
# 7) K-FOLD: BUILD SPLITS -> TRAIN -> EVAL -> AGGREGATE
# -----
def build_kfold_splits(df_std, n_splits=7, seed=42, val_within_remainder=0.1765):

```

```

"""
For each fold:
    - Test = 1/n_splits of data (stratified by label) ~15%
    - From the remainder (~85%), Val = val_within_remainder of remainder (so over
    - Train = the rest (~70%)
Returns a list of fold DataFrames with 'split' = Train/Val/Test.
"""

df = df_std.copy().reset_index(drop=True)
y = df['label'].astype(str).values
idx = np.arange(len(df))
skf = StratifiedKFold(n_splits=n_splits, shuffle=True, random_state=seed)

folds = []
for fold_id, (trainval_idx, test_idx) in enumerate(skf.split(idx, y), 1):
    rest_labels = y[trainval_idx]
    sss2 = StratifiedShuffleSplit(n_splits=1, test_size=val_within_remainder, r
    for tr_rel, val_rel in sss2.split(trainval_idx, rest_labels):
        tr_idx = trainval_idx[tr_rel]
        val_idx = trainval_idx[val_rel]

    split = pd.Series(index=np.arange(len(df)), data="Ignore", dtype=str)
    split.loc[tr_idx] = "Train"
    split.loc[val_idx] = "Val"
    split.loc[test_idx] = "Test"

    dfi = df.copy()
    dfi['split'] = split.values
    dfi = dfi[dfi['split'].isin(['Train', 'Val', 'Test'])].reset_index(drop=True)

    # Quick fold report
    nT = (dfi.split=='Train').sum(); nV = (dfi.split=='Val').sum(); nE = (dfi.s
    print(f"Fold {fold_id}: Train={nT} ({100*nT/len(dfi):.1f}%) Val={nV} ({100
    folds.append(dfi)
return folds

def run_one_fold(fold_df, fold_id, config):
    print("\n" + "#" * 88)
    print(f"### FOLD {fold_id} ### Train={sum(fold_df.split=='Train')}, Val={sum(f
    print("#" * 88)

    dls, sizes, class_names, meta_vocab = make_dataloaders_from_df(config, fold_df)

    # --- MODEL INSTANTIATION CHANGED ---
    model = ResNet50Multimodal(num_classes=len(class_names),
                               meta_vocab=meta_vocab,
                               meta_hidden_dim=config.META_HIDDEN_DIM,
                               meta_out_dim=config.META_OUT_DIM,
                               meta_dropout=config.META_DROPOUT).to(config.DEVICE)

    # Optional model size / FLOPs
    total_params = sum(p.numel() for p in model.parameters())
    trainable_params = sum(p.numel() for p in model.parameters() if p.requires_grad
    print(f"Params: {total_params/1e6:.2f}M | Trainable: {trainable_params/1e6:.2f}
    if HAVE_THOP:
        try:
            dummy_img = torch.randn(1,3,config.IMG_SIZE,config.IMG_SIZE).to(config.

```

```

        dummy_meta = {
            'age_group': torch.zeros(1, dtype=torch.long, device=config.DEVICE)
            'gender':    torch.zeros(1, dtype=torch.long, device=config.DEVICE)
            'skin_tone': torch.zeros(1, dtype=torch.long, device=config.DEVICE)
        }
        macs, params = profile(model, inputs=(dummy_img, dummy_meta), verbose=False)
        print(f"Approx MACs: {macs/1e9:.2f} G, Params (THOP): {params/1e6:.2f} M")
    except Exception as e:
        print("THOP profile failed (skipped):", e)

model, _ = train_model(model, dls, sizes, num_epochs=config.EPOCHS)

best_path = os.path.join(config.RESULTS_DIR, 'best_multimodal_resnet.pth')
if os.path.exists(best_path):
    model.load_state_dict(torch.load(best_path, map_location=config.DEVICE))

metrics = evaluate_and_visualize(model, dls['Test'], class_names, fold_df)
return metrics

def kfold_cross_val(df_std, config):
    folds = build_kfold_splits(df_std,
                               n_splits=config.N_SPLITS,
                               seed=42,
                               val_within_remainder=config.VAL_WITHIN_REMAINDER)

    per_fold = []
    for i, fold_df in enumerate(folds, 1):
        m = run_one_fold(fold_df, i, config)
        m['fold'] = i
        per_fold.append(m)

    perf = pd.DataFrame(per_fold).set_index('fold')
    def mean_std(col):
        vals = perf[col].astype(float).values
        return float(np.nanmean(vals)), float(np.nanstd(vals))

    summary = {
        'acc_mean': mean_std('accuracy')[0], 'acc_std': mean_std('accuracy')[1],
        'roc_macro_mean': mean_std('roc_auc_macro')[0], 'roc_macro_std': mean_std('roc_auc_macro')[1],
        'kappa_mean': mean_std('kappa')[0], 'kappa_std': mean_std('kappa')[1],
        'mcc_mean': mean_std('mcc')[0], 'mcc_std': mean_std('mcc')[1],
        'wga_all_mean': mean_std('wga_all')[0], 'wga_all_std': mean_std('wga_all')[1],
        'eod_all_mean': mean_std('eod_all')[0], 'eod_all_std': mean_std('eod_all')[1],
        'wga_filt_mean': mean_std('wga_filtered')[0], 'wga_filt_std': mean_std('wga_filtered')[1],
        'eod_filt_mean': mean_std('eod_filtered')[0], 'eod_filt_std': mean_std('eod_filtered')[1]
    }

    _print_section("OVERALL K-FOLD RESULTS")
    print(f"Accuracy          : {summary['acc_mean']:.4f} ± {summary['acc_std']:.4f}")
    print(f"ROC AUC (Macro)     : {summary['roc_macro_mean']:.4f} ± {summary['roc_macro_std']:.4f}")
    print(f"Cohen's Kappa       : {summary['kappa_mean']:.4f} ± {summary['kappa_std']:.4f}")
    print(f"MCC                 : {summary['mcc_mean']:.4f} ± {summary['mcc_std']:.4f}")
    print("-"*40)
    print(f"WGA (All Groups)    : {summary['wga_all_mean']:.4f} ± {summary['wga_all_std']:.4f}")
    print(f"EOD (All Groups)    : {summary['eod_all_mean']:.4f} ± {summary['eod_all_std']:.4f}")
    print(f"WGA (n>={config.MIN_SUPPORT}) : {summary['wga_filt_mean']:.4f} ± {summary['wga_filt_std']:.4f}")
    print(f"EOD (n>={config.MIN_SUPPORT}) : {summary['eod_filt_mean']:.4f} ± {summary['eod_filt_std']:.4f}")

```

```

print()

# ---- POOLED SUBGROUP ACCURACY TABLE ----
pooled_counts = defaultdict(lambda: {'correct': 0, 'total': 0})
for _, row in perf.iterrows():
    for group_type, group_data in row['pooled_groups'].items():
        for group_name, (correct, total) in group_data.items():
            key = f"{group_type}:{group_name}"
            pooled_counts[key]['correct'] += correct
            pooled_counts[key]['total'] += total

pooled_results = []
for key, data in pooled_counts.items():
    if data['total'] >= config.MIN_SUPPORT:
        acc = data['correct'] / data['total'] if data['total'] > 0 else 0
        pooled_results.append({
            'Group': key,
            'Accuracy': acc,
            'Support': data['total']
        })

df_pooled = pd.DataFrame(pooled_results).sort_values(by=['Group']).reset_index()
if not df_pooled.empty:
    _print_section(f"POOLED SUBGROUP ACCURACY ACROSS ALL FOLDS (SUPPORT >= {config.MIN_SUPPORT})")
    print(df_pooled.to_string())

# Save summary and per-fold results
with open(os.path.join(config.RESULTS_DIR, "summary_results.json"), "w") as f:
    json.dump(summary, f, indent=4)
perf.drop(columns=['pooled_groups']).to_csv(os.path.join(config.RESULTS_DIR, "per_fold_results.csv"), index=False)
if not df_pooled.empty:
    df_pooled.to_csv(os.path.join(config.RESULTS_DIR, "pooled_subgroup_accuracy.csv"), index=False)

print(f"\nResults saved to '{config.RESULTS_DIR}'")
return summary, perf, df_pooled

# -----
# 8) MAIN EXECUTION
# -----
if __name__ == '__main__':
    try:
        df_orig = pd.read_csv(config.ORIGINAL_METADATA_CSV)
        df_std = standardize_metadata_columns(df_orig)
        print(f"Loaded and standardized metadata: {len(df_std)} rows.")

        summary, per_fold_results, pooled_df = kfold_cross_val(df_std, config)

    except FileNotFoundError:
        print(f"ERROR: Metadata file not found at '{config.ORIGINAL_METADATA_CSV}'")
        print("Please update the 'ORIGINAL_METADATA_CSV' path in the Config class.")
    except Exception as e:
        print(f"An unexpected error occurred: {e}")
        import traceback
        traceback.print_exc()

```

Device: cuda
Model: ResNet-50
K-Fold: N_SPLITS=7 (~Test 14.3%), Val_in_remainder=0.175
Loaded and standardized metadata: 565 rows.
Fold 1: Train=399 (70.6%) Val=85 (15.0%) Test=81 (14.3%)
Fold 2: Train=399 (70.6%) Val=85 (15.0%) Test=81 (14.3%)
Fold 3: Train=399 (70.6%) Val=85 (15.0%) Test=81 (14.3%)
Fold 4: Train=399 (70.6%) Val=85 (15.0%) Test=81 (14.3%)
Fold 5: Train=399 (70.6%) Val=85 (15.0%) Test=81 (14.3%)
Fold 6: Train=400 (70.8%) Val=85 (15.0%) Test=80 (14.2%)
Fold 7: Train=400 (70.8%) Val=85 (15.0%) Test=80 (14.2%)

FOLD 1 ### Train=399, Val=85, Test=81

Params: 24.73M | Trainable: 1.22M (4.93%)
Approx MACs: 4.13 G, Params (THOP): 24.73 M

=====
START TRAINING (Multimodal ResNet-50) for 200 epochs
=====

Epoch 1/200	Train 1.6921/0.2206	Val 1.6023/0.2706	LR 2.00e-05
Epoch 2/200	Train 1.6355/0.2581	Val 1.5896/0.3882	LR 3.00e-05
Epoch 3/200	Train 1.5444/0.3308	Val 1.5657/0.4588	LR 4.00e-05
Epoch 4/200	Train 1.4499/0.4561	Val 1.5320/0.5176	LR 5.00e-05
Epoch 5/200	Train 1.3657/0.5113	Val 1.4723/0.5882	LR 6.00e-05
Epoch 6/200	Train 1.2554/0.6090	Val 1.3893/0.7059	LR 7.00e-05
Epoch 7/200	Train 1.1316/0.6992	Val 1.2925/0.7412	LR 8.00e-05
Epoch 8/200	Train 1.0775/0.7043	Val 1.1917/0.7412	LR 9.00e-05
Epoch 9/200	Train 0.9869/0.7995	Val 1.0930/0.7529	LR 1.00e-04
Epoch 10/200	Train 0.9185/0.7945	Val 1.0133/0.7529	LR 1.00e-04
Epoch 11/200	Train 0.8916/0.8045	Val 0.9533/0.7882	LR 1.00e-04
Epoch 12/200	Train 0.8356/0.8195	Val 0.9120/0.7882	LR 1.00e-04
Epoch 13/200	Train 0.7915/0.8396	Val 0.8863/0.7882	LR 9.99e-05
Epoch 14/200	Train 0.7773/0.8446	Val 0.8656/0.8118	LR 9.99e-05
Epoch 15/200	Train 0.7334/0.8672	Val 0.8543/0.8118	LR 9.98e-05
Epoch 16/200	Train 0.7217/0.8797	Val 0.8392/0.8118	LR 9.98e-05
Epoch 17/200	Train 0.7216/0.8747	Val 0.8288/0.8118	LR 9.97e-05
Epoch 18/200	Train 0.6875/0.9023	Val 0.8168/0.8118	LR 9.96e-05
Epoch 19/200	Train 0.6763/0.8897	Val 0.8023/0.8235	LR 9.94e-05
Epoch 20/200	Train 0.6735/0.8897	Val 0.7890/0.8353	LR 9.93e-05
Epoch 21/200	Train 0.6517/0.9048	Val 0.7763/0.8353	LR 9.92e-05
Epoch 22/200	Train 0.6495/0.9198	Val 0.7682/0.8353	LR 9.90e-05
Epoch 23/200	Train 0.6437/0.9173	Val 0.7624/0.8235	LR 9.88e-05
Epoch 24/200	Train 0.6245/0.9073	Val 0.7557/0.8235	LR 9.87e-05
Epoch 25/200	Train 0.6136/0.9248	Val 0.7469/0.8235	LR 9.85e-05
Epoch 26/200	Train 0.5975/0.9373	Val 0.7504/0.8235	LR 9.83e-05
Epoch 27/200	Train 0.5978/0.9373	Val 0.7416/0.8353	LR 9.80e-05
Epoch 28/200	Train 0.5901/0.9373	Val 0.7377/0.8353	LR 9.78e-05
Epoch 29/200	Train 0.5895/0.9348	Val 0.7378/0.8353	LR 9.76e-05
Epoch 30/200	Train 0.5629/0.9499	Val 0.7291/0.8353	LR 9.73e-05
Epoch 31/200	Train 0.5791/0.9348	Val 0.7196/0.8471	LR 9.70e-05
Epoch 32/200	Train 0.5612/0.9474	Val 0.7177/0.8471	LR 9.67e-05
Epoch 33/200	Train 0.5714/0.9348	Val 0.7130/0.8471	LR 9.64e-05

Epoch 34/200	Train 0.5522/0.9373	Val 0.7065/0.8471	LR 9.61e-05
Epoch 35/200	Train 0.5624/0.9323	Val 0.7035/0.8471	LR 9.58e-05
Epoch 36/200	Train 0.5638/0.9398	Val 0.7000/0.8471	LR 9.55e-05
Epoch 37/200	Train 0.5392/0.9524	Val 0.6941/0.8471	LR 9.51e-05
Epoch 38/200	Train 0.5411/0.9674	Val 0.6872/0.8588	LR 9.47e-05
Epoch 39/200	Train 0.5529/0.9624	Val 0.6820/0.8588	LR 9.44e-05
Epoch 40/200	Train 0.5471/0.9524	Val 0.6813/0.8588	LR 9.40e-05
Epoch 41/200	Train 0.5390/0.9524	Val 0.6804/0.8588	LR 9.36e-05
Epoch 42/200	Train 0.5253/0.9624	Val 0.6774/0.8588	LR 9.32e-05
Epoch 43/200	Train 0.5454/0.9499	Val 0.6739/0.8588	LR 9.27e-05
Epoch 44/200	Train 0.5343/0.9549	Val 0.6747/0.8588	LR 9.23e-05
Epoch 45/200	Train 0.5299/0.9549	Val 0.6718/0.8588	LR 9.19e-05
Epoch 46/200	Train 0.5188/0.9674	Val 0.6752/0.8471	LR 9.14e-05
Epoch 47/200	Train 0.5161/0.9699	Val 0.6771/0.8471	LR 9.09e-05
Epoch 48/200	Train 0.5291/0.9549	Val 0.6714/0.8588	LR 9.05e-05
Epoch 49/200	Train 0.5241/0.9624	Val 0.6608/0.8706	LR 9.00e-05
Epoch 50/200	Train 0.5328/0.9474	Val 0.6601/0.8706	LR 8.95e-05
Epoch 51/200	Train 0.5262/0.9549	Val 0.6630/0.8588	LR 8.89e-05
Epoch 52/200	Train 0.5032/0.9724	Val 0.6630/0.8588	LR 8.84e-05
Epoch 53/200	Train 0.5139/0.9624	Val 0.6568/0.8588	LR 8.79e-05
Epoch 54/200	Train 0.5129/0.9549	Val 0.6549/0.8588	LR 8.73e-05
Epoch 55/200	Train 0.5148/0.9624	Val 0.6513/0.8588	LR 8.68e-05
Epoch 56/200	Train 0.5170/0.9624	Val 0.6444/0.8588	LR 8.62e-05
Epoch 57/200	Train 0.5166/0.9574	Val 0.6406/0.8706	LR 8.56e-05
Epoch 58/200	Train 0.5139/0.9599	Val 0.6352/0.8824	LR 8.51e-05
Epoch 59/200	Train 0.5024/0.9799	Val 0.6323/0.8941	LR 8.45e-05
Epoch 60/200	Train 0.5098/0.9699	Val 0.6271/0.8941	LR 8.39e-05
Epoch 61/200	Train 0.4912/0.9774	Val 0.6286/0.8941	LR 8.33e-05
Epoch 62/200	Train 0.4931/0.9799	Val 0.6295/0.8824	LR 8.26e-05
Epoch 63/200	Train 0.4909/0.9674	Val 0.6301/0.8941	LR 8.20e-05
Epoch 64/200	Train 0.4970/0.9774	Val 0.6303/0.8824	LR 8.14e-05
Epoch 65/200	Train 0.4884/0.9724	Val 0.6300/0.8824	LR 8.07e-05
Epoch 66/200	Train 0.4859/0.9774	Val 0.6254/0.8824	LR 8.01e-05
Epoch 67/200	Train 0.4952/0.9749	Val 0.6278/0.8824	LR 7.94e-05
Epoch 68/200	Train 0.4869/0.9674	Val 0.6357/0.8824	LR 7.87e-05
Epoch 69/200	Train 0.4811/0.9699	Val 0.6352/0.8824	LR 7.80e-05
Epoch 70/200	Train 0.4807/0.9774	Val 0.6342/0.8824	LR 7.73e-05
Epoch 71/200	Train 0.4906/0.9774	Val 0.6314/0.8824	LR 7.67e-05
Epoch 72/200	Train 0.4790/0.9850	Val 0.6342/0.8824	LR 7.59e-05
Epoch 73/200	Train 0.4841/0.9774	Val 0.6271/0.8941	LR 7.52e-05
Epoch 74/200	Train 0.4723/0.9950	Val 0.6276/0.8941	LR 7.45e-05
Epoch 75/200	Train 0.4922/0.9749	Val 0.6267/0.8941	LR 7.38e-05
Epoch 76/200	Train 0.4873/0.9774	Val 0.6276/0.8941	LR 7.31e-05
Epoch 77/200	Train 0.4752/0.9875	Val 0.6276/0.9059	LR 7.23e-05
Epoch 78/200	Train 0.4721/0.9850	Val 0.6252/0.9059	LR 7.16e-05
Epoch 79/200	Train 0.4738/0.9799	Val 0.6212/0.9059	LR 7.08e-05
Epoch 80/200	Train 0.4799/0.9850	Val 0.6254/0.8941	LR 7.01e-05
Epoch 81/200	Train 0.4758/0.9875	Val 0.6245/0.9059	LR 6.93e-05
Epoch 82/200	Train 0.4759/0.9749	Val 0.6216/0.9059	LR 6.86e-05
Epoch 83/200	Train 0.4734/0.9900	Val 0.6194/0.9059	LR 6.78e-05
Epoch 84/200	Train 0.4758/0.9875	Val 0.6178/0.9059	LR 6.70e-05
Epoch 85/200	Train 0.4823/0.9799	Val 0.6163/0.9059	LR 6.62e-05
Epoch 86/200	Train 0.4799/0.9825	Val 0.6164/0.8941	LR 6.55e-05
Epoch 87/200	Train 0.4795/0.9799	Val 0.6162/0.9059	LR 6.47e-05
Epoch 88/200	Train 0.4716/0.9799	Val 0.6174/0.9059	LR 6.39e-05
Epoch 89/200	Train 0.4729/0.9900	Val 0.6157/0.9059	LR 6.31e-05

```

Epoch 90/200 | Train 0.4721/0.9875 | Val 0.6176/0.9059 | LR 6.23e-05
Epoch 91/200 | Train 0.4661/0.9850 | Val 0.6149/0.9059 | LR 6.15e-05
Epoch 92/200 | Train 0.4597/0.9900 | Val 0.6146/0.9059 | LR 6.07e-05
Epoch 93/200 | Train 0.4684/0.9850 | Val 0.6123/0.9059 | LR 5.99e-05
Epoch 94/200 | Train 0.4746/0.9875 | Val 0.6100/0.9059 | LR 5.90e-05
Epoch 95/200 | Train 0.4674/0.9875 | Val 0.6123/0.9059 | LR 5.82e-05
Epoch 96/200 | Train 0.4728/0.9925 | Val 0.6181/0.9059 | LR 5.74e-05
Epoch 97/200 | Train 0.4740/0.9850 | Val 0.6166/0.9059 | LR 5.66e-05
Epoch 98/200 | Train 0.4669/0.9875 | Val 0.6202/0.9059 | LR 5.58e-05
Epoch 99/200 | Train 0.4570/0.9925 | Val 0.6234/0.9059 | LR 5.50e-05
Epoch 100/200 | Train 0.4668/0.9799 | Val 0.6234/0.9059 | LR 5.41e-05
Epoch 101/200 | Train 0.4715/0.9825 | Val 0.6286/0.9059 | LR 5.33e-05
Epoch 102/200 | Train 0.4633/0.9900 | Val 0.6288/0.9059 | LR 5.25e-05
Epoch 103/200 | Train 0.4658/0.9850 | Val 0.6256/0.9059 | LR 5.17e-05
Epoch 104/200 | Train 0.4843/0.9850 | Val 0.6344/0.8824 | LR 5.08e-05
Epoch 105/200 | Train 0.4646/0.9900 | Val 0.6300/0.8941 | LR 5.00e-05
Epoch 106/200 | Train 0.4618/0.9900 | Val 0.6271/0.9059 | LR 4.92e-05
Epoch 107/200 | Train 0.4627/0.9925 | Val 0.6273/0.9059 | LR 4.83e-05
Epoch 108/200 | Train 0.4549/0.9950 | Val 0.6232/0.8941 | LR 4.75e-05
Epoch 109/200 | Train 0.4647/0.9850 | Val 0.6238/0.8941 | LR 4.67e-05
Epoch 110/200 | Train 0.4578/0.9925 | Val 0.6219/0.8941 | LR 4.59e-05
Epoch 111/200 | Train 0.4527/0.9900 | Val 0.6189/0.8941 | LR 4.50e-05
Epoch 112/200 | Train 0.4613/0.9850 | Val 0.6178/0.8941 | LR 4.42e-05
Epoch 113/200 | Train 0.4543/0.9950 | Val 0.6200/0.8941 | LR 4.34e-05
Epoch 114/200 | Train 0.4647/0.9875 | Val 0.6182/0.8941 | LR 4.26e-05
Early stopping.
Training complete in 6.4 minutes. Best Val Loss 0.6100

```

=====

OVERALL PERFORMANCE (ON COMBINED DATASET)

=====

	precision	recall	f1-score	support
Herpes	1.00	0.29	0.44	7
Melanoma	0.75	1.00	0.86	3
Monkeypox	1.00	0.89	0.94	18
Sarampion	0.73	0.83	0.77	29
Varicela	0.77	0.83	0.80	24
accuracy			0.80	81
macro avg	0.85	0.77	0.76	81
weighted avg	0.82	0.80	0.79	81

=====

PERFORMANCE ANALYSIS BY: DOMINANT_GENDER

=====

--- Results for Group: Man ---

	precision	recall	f1-score	support
Herpes	1.00	0.50	0.67	4
Melanoma	0.75	1.00	0.86	3
Monkeypox	1.00	0.88	0.93	16
Sarampion	0.72	0.76	0.74	17

Varicela	0.69	0.79	0.73	14
accuracy			0.80	54
macro avg	0.83	0.79	0.79	54
weighted avg	0.82	0.80	0.80	54

--- Results for Group: Woman ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	3
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	0.73	0.92	0.81	12
Varicela	0.90	0.90	0.90	10
accuracy			0.81	27
macro avg	0.53	0.56	0.54	27
weighted avg	0.73	0.81	0.77	27

=====

PERFORMANCE ANALYSIS BY: AGE_GROUP

=====

--- Results for Group: 0-17 ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	0.60	0.75	0.67	4
Varicela	0.75	0.75	0.75	4
accuracy			0.73	11
macro avg	0.47	0.50	0.48	11
weighted avg	0.67	0.73	0.70	11

--- Results for Group: 18-59 ---

	precision	recall	f1-score	support
Herpes	1.00	0.33	0.50	6
Melanoma	0.75	1.00	0.86	3
Monkeypox	1.00	0.88	0.93	16
Sarampion	0.75	0.84	0.79	25
Varicela	0.77	0.85	0.81	20
accuracy			0.81	70
macro avg	0.85	0.78	0.78	70
weighted avg	0.84	0.81	0.81	70

=====

PERFORMANCE ANALYSIS BY: DOMINANT_RACE

=====

--- Results for Group: asian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	0.50	1.00	0.67	1
Varicela	1.00	1.00	1.00	4
accuracy			0.89	9
macro avg	0.50	0.60	0.53	9
weighted avg	0.83	0.89	0.85	9

--- Results for Group: black ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	0.86	0.92	7
Sarampion	0.50	1.00	0.67	1
Varicela	0.00	0.00	0.00	0
accuracy			0.89	9
macro avg	0.50	0.57	0.52	9
weighted avg	0.94	0.89	0.90	9

--- Results for Group: indian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	0.80	1.00	0.89	4
Varicela	0.00	0.00	0.00	0
accuracy			0.88	8
macro avg	0.36	0.40	0.38	8
weighted avg	0.78	0.88	0.82	8

--- Results for Group: latino hispanic ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	1.00	1.00	1.00	4
Varicela	0.00	0.00	0.00	0
accuracy			1.00	5
macro avg	0.40	0.40	0.40	5

weighted avg	1.00	1.00	1.00	5
--------------	------	------	------	---

--- Results for Group: middle eastern ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	0.00	0.00	0.00	0
Varicela	0.00	0.00	0.00	0
accuracy			1.00	1
macro avg	0.20	0.20	0.20	1
weighted avg	1.00	1.00	1.00	1

--- Results for Group: white ---

	precision	recall	f1-score	support
Herpes	1.00	0.25	0.40	4
Melanoma	0.75	1.00	0.86	3
Monkeypox	1.00	0.67	0.80	3
Sarampion	0.70	0.74	0.72	19
Varicela	0.73	0.80	0.76	20
accuracy			0.73	49
macro avg	0.84	0.69	0.71	49
weighted avg	0.76	0.73	0.72	49

=====

FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.0000 EOD (max TPR gap): 1.0000

Worst pair: 0-17|female|asian vs 18-59|male|black

=====

=====

FAIRNESS SUMMARY (FILTERED, n >= 15) | WGA: 0.6522 EOD (max TPR gap): 0.3499

Worst pair (among filtered): 18-59|female|white vs 18-59|male|white

Groups used (filtered): ['18-59|female|white', '18-59|male|white']

=====

#####

####

FOLD 2 ### Train=399, Val=85, Test=81

#####

####

Params: 24.73M | Trainable: 1.22M (4.93%)

Approx MACs: 4.13 G, Params (THOP): 24.73 M

=====

START TRAINING (Multimodal ResNet-50) for 200 epochs

=====

Epoch 1/200 | Train 1.7936/0.1704 | Val 1.6218/0.0706 | LR 2.00e-05

Epoch 2/200 | Train 1.7507/0.1830 | Val 1.6078/0.1059 | LR 3.00e-05

Epoch 3/200	Train 1.6819/0.2356	Val 1.5820/0.3059	LR 4.00e-05
Epoch 4/200	Train 1.5577/0.3283	Val 1.5399/0.4706	LR 5.00e-05
Epoch 5/200	Train 1.4067/0.4386	Val 1.4773/0.5882	LR 6.00e-05
Epoch 6/200	Train 1.2754/0.5539	Val 1.3957/0.6353	LR 7.00e-05
Epoch 7/200	Train 1.1807/0.6165	Val 1.2953/0.6941	LR 8.00e-05
Epoch 8/200	Train 1.0743/0.7143	Val 1.1919/0.7176	LR 9.00e-05
Epoch 9/200	Train 1.0013/0.7444	Val 1.1003/0.7059	LR 1.00e-04
Epoch 10/200	Train 0.9548/0.7594	Val 1.0275/0.7176	LR 1.00e-04
Epoch 11/200	Train 0.8740/0.8170	Val 0.9792/0.7412	LR 1.00e-04
Epoch 12/200	Train 0.8482/0.7945	Val 0.9480/0.7529	LR 1.00e-04
Epoch 13/200	Train 0.8089/0.8195	Val 0.9228/0.7529	LR 9.99e-05
Epoch 14/200	Train 0.7870/0.8421	Val 0.9031/0.7529	LR 9.99e-05
Epoch 15/200	Train 0.7624/0.8446	Val 0.8908/0.7765	LR 9.98e-05
Epoch 16/200	Train 0.7460/0.8471	Val 0.8927/0.7765	LR 9.98e-05
Epoch 17/200	Train 0.7128/0.8697	Val 0.8857/0.7882	LR 9.97e-05
Epoch 18/200	Train 0.7050/0.8947	Val 0.8884/0.7882	LR 9.96e-05
Epoch 19/200	Train 0.6938/0.8847	Val 0.8637/0.8000	LR 9.94e-05
Epoch 20/200	Train 0.6731/0.8997	Val 0.8417/0.7882	LR 9.93e-05
Epoch 21/200	Train 0.6570/0.9048	Val 0.8278/0.7882	LR 9.92e-05
Epoch 22/200	Train 0.6521/0.9023	Val 0.8181/0.7882	LR 9.90e-05
Epoch 23/200	Train 0.6440/0.9148	Val 0.8118/0.7882	LR 9.88e-05
Epoch 24/200	Train 0.6393/0.9123	Val 0.8066/0.7765	LR 9.87e-05
Epoch 25/200	Train 0.6058/0.9198	Val 0.8034/0.7765	LR 9.85e-05
Epoch 26/200	Train 0.6186/0.9273	Val 0.8034/0.7765	LR 9.83e-05
Epoch 27/200	Train 0.6247/0.9223	Val 0.7931/0.7765	LR 9.80e-05
Epoch 28/200	Train 0.5989/0.9223	Val 0.7906/0.7765	LR 9.78e-05
Epoch 29/200	Train 0.5944/0.9398	Val 0.7874/0.7882	LR 9.76e-05
Epoch 30/200	Train 0.6023/0.9248	Val 0.7857/0.7882	LR 9.73e-05
Epoch 31/200	Train 0.5751/0.9398	Val 0.7736/0.8118	LR 9.70e-05
Epoch 32/200	Train 0.5807/0.9449	Val 0.7726/0.7882	LR 9.67e-05
Epoch 33/200	Train 0.5709/0.9398	Val 0.7693/0.8118	LR 9.64e-05
Epoch 34/200	Train 0.5729/0.9398	Val 0.7654/0.8118	LR 9.61e-05
Epoch 35/200	Train 0.5809/0.9398	Val 0.7666/0.8000	LR 9.58e-05
Epoch 36/200	Train 0.5601/0.9499	Val 0.7631/0.8118	LR 9.55e-05
Epoch 37/200	Train 0.5546/0.9474	Val 0.7605/0.8000	LR 9.51e-05
Epoch 38/200	Train 0.5398/0.9624	Val 0.7595/0.8118	LR 9.47e-05
Epoch 39/200	Train 0.5480/0.9499	Val 0.7563/0.8353	LR 9.44e-05
Epoch 40/200	Train 0.5395/0.9549	Val 0.7477/0.8235	LR 9.40e-05
Epoch 41/200	Train 0.5363/0.9599	Val 0.7458/0.8235	LR 9.36e-05
Epoch 42/200	Train 0.5213/0.9699	Val 0.7453/0.8235	LR 9.32e-05
Epoch 43/200	Train 0.5290/0.9574	Val 0.7437/0.8118	LR 9.27e-05
Epoch 44/200	Train 0.5313/0.9599	Val 0.7426/0.8235	LR 9.23e-05
Epoch 45/200	Train 0.5295/0.9524	Val 0.7434/0.8235	LR 9.19e-05
Epoch 46/200	Train 0.5297/0.9599	Val 0.7427/0.8353	LR 9.14e-05
Epoch 47/200	Train 0.5133/0.9674	Val 0.7397/0.8235	LR 9.09e-05
Epoch 48/200	Train 0.5206/0.9699	Val 0.7394/0.8235	LR 9.05e-05
Epoch 49/200	Train 0.5208/0.9574	Val 0.7343/0.8235	LR 9.00e-05
Epoch 50/200	Train 0.5171/0.9599	Val 0.7341/0.8471	LR 8.95e-05
Epoch 51/200	Train 0.5218/0.9549	Val 0.7297/0.8471	LR 8.89e-05
Epoch 52/200	Train 0.5149/0.9524	Val 0.7302/0.8471	LR 8.84e-05
Epoch 53/200	Train 0.5031/0.9774	Val 0.7274/0.8471	LR 8.79e-05
Epoch 54/200	Train 0.4993/0.9799	Val 0.7297/0.8471	LR 8.73e-05
Epoch 55/200	Train 0.5034/0.9699	Val 0.7261/0.8471	LR 8.68e-05
Epoch 56/200	Train 0.4976/0.9749	Val 0.7177/0.8353	LR 8.62e-05
Epoch 57/200	Train 0.4935/0.9749	Val 0.7164/0.8353	LR 8.56e-05
Epoch 58/200	Train 0.5001/0.9724	Val 0.7137/0.8353	LR 8.51e-05

```

Epoch 59/200 | Train 0.4969/0.9674 | Val 0.7182/0.8353 | LR 8.45e-05
Epoch 60/200 | Train 0.4988/0.9774 | Val 0.7204/0.8353 | LR 8.39e-05
Epoch 61/200 | Train 0.4888/0.9825 | Val 0.7286/0.8353 | LR 8.33e-05
Epoch 62/200 | Train 0.4855/0.9900 | Val 0.7181/0.8235 | LR 8.26e-05
Epoch 63/200 | Train 0.4925/0.9749 | Val 0.7244/0.8235 | LR 8.20e-05
Epoch 64/200 | Train 0.4925/0.9799 | Val 0.7239/0.8353 | LR 8.14e-05
Epoch 65/200 | Train 0.5019/0.9624 | Val 0.7224/0.8353 | LR 8.07e-05
Epoch 66/200 | Train 0.4942/0.9574 | Val 0.7313/0.8353 | LR 8.01e-05
Epoch 67/200 | Train 0.4902/0.9774 | Val 0.7355/0.8235 | LR 7.94e-05
Epoch 68/200 | Train 0.4952/0.9774 | Val 0.7306/0.8235 | LR 7.87e-05
Epoch 69/200 | Train 0.4874/0.9724 | Val 0.7288/0.8235 | LR 7.80e-05
Epoch 70/200 | Train 0.4815/0.9825 | Val 0.7379/0.8235 | LR 7.73e-05
Epoch 71/200 | Train 0.4902/0.9799 | Val 0.7410/0.8235 | LR 7.67e-05
Epoch 72/200 | Train 0.4759/0.9900 | Val 0.7390/0.8118 | LR 7.59e-05
Epoch 73/200 | Train 0.4796/0.9850 | Val 0.7382/0.8353 | LR 7.52e-05
Epoch 74/200 | Train 0.4826/0.9774 | Val 0.7322/0.8471 | LR 7.45e-05
Epoch 75/200 | Train 0.4832/0.9799 | Val 0.7298/0.8353 | LR 7.38e-05
Epoch 76/200 | Train 0.4672/0.9875 | Val 0.7196/0.8471 | LR 7.31e-05
Epoch 77/200 | Train 0.4828/0.9724 | Val 0.7170/0.8471 | LR 7.23e-05
Epoch 78/200 | Train 0.4791/0.9850 | Val 0.7187/0.8353 | LR 7.16e-05

```

Early stopping.

Training complete in 4.5 minutes. Best Val Loss 0.7137

=====

OVERALL PERFORMANCE (ON COMBINED DATASET)

=====

	precision	recall	f1-score	support
Herpes	0.75	0.86	0.80	7
Melanoma	1.00	0.67	0.80	3
Monkeypox	1.00	0.83	0.91	18
Sarampion	0.81	0.86	0.83	29
Varicela	0.84	0.88	0.86	24
accuracy			0.85	81
macro avg	0.88	0.82	0.84	81
weighted avg	0.86	0.85	0.85	81

=====

PERFORMANCE ANALYSIS BY: DOMINANT_GENDER

=====

--- Results for Group: Man ---

	precision	recall	f1-score	support
Herpes	0.67	0.80	0.73	5
Melanoma	1.00	0.67	0.80	3
Monkeypox	1.00	0.93	0.96	14
Sarampion	0.73	0.73	0.73	15
Varicela	0.75	0.82	0.78	11
accuracy			0.81	48
macro avg	0.83	0.79	0.80	48
weighted avg	0.82	0.81	0.82	48

--- Results for Group: Woman ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	2
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	0.50	0.67	4
Sarampion	0.88	1.00	0.93	14
Varicela	0.92	0.92	0.92	13
accuracy			0.91	33
macro avg	0.76	0.68	0.70	33
weighted avg	0.92	0.91	0.90	33

=====

PERFORMANCE ANALYSIS BY: AGE_GROUP

=====

--- Results for Group: 0-17 ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	1
Sarampion	1.00	0.33	0.50	3
Varicela	0.57	1.00	0.73	4
accuracy			0.62	8
macro avg	0.31	0.27	0.25	8
weighted avg	0.66	0.62	0.55	8

--- Results for Group: 18-59 ---

	precision	recall	f1-score	support
Herpes	0.71	0.83	0.77	6
Melanoma	1.00	0.67	0.80	3
Monkeypox	1.00	0.88	0.94	17
Sarampion	0.80	0.92	0.86	26
Varicela	0.94	0.85	0.89	20
accuracy			0.88	72
macro avg	0.89	0.83	0.85	72
weighted avg	0.89	0.88	0.88	72

--- Results for Group: 60+ ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	0.00	0.00	0.00	0

Varicela	0.00	0.00	0.00	0
accuracy			1.00	1
macro avg	0.20	0.20	0.20	1
weighted avg	1.00	1.00	1.00	1

```
=====
PERFORMANCE ANALYSIS BY: DOMINANT_RACE
=====
```

```
--- Results for Group: asian ---
      precision    recall  f1-score   support

   Herpes         1.00      0.67      0.80         3
   Melanoma        0.00      0.00      0.00         0
 Monkeypox         1.00      1.00      1.00         4
 Sarampion         0.67      1.00      0.80         2
   Varicela         1.00      1.00      1.00         2

 accuracy                   0.91         11
 macro avg              0.73      0.73      0.72         11
 weighted avg           0.94      0.91      0.91         11
```

```
--- Results for Group: black ---
      precision    recall  f1-score   support

   Herpes         0.00      0.00      0.00         0
   Melanoma        0.00      0.00      0.00         0
 Monkeypox         1.00      1.00      1.00         2
 Sarampion         0.00      0.00      0.00         0
   Varicela         0.00      0.00      0.00         0

 accuracy                   1.00         2
 macro avg              0.20      0.20      0.20         2
 weighted avg           1.00      1.00      1.00         2
```

```
--- Results for Group: indian ---
      precision    recall  f1-score   support

   Herpes         0.00      0.00      0.00         0
   Melanoma        0.00      0.00      0.00         0
 Monkeypox         1.00      1.00      1.00         2
 Sarampion         1.00      0.50      0.67         2
   Varicela         0.00      0.00      0.00         0

 accuracy                   0.75         4
 macro avg              0.40      0.30      0.33         4
 weighted avg           1.00      0.75      0.83         4
```

```
--- Results for Group: latino hispanic ---
      precision    recall  f1-score   support
```

Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	0.67	0.80	3
Sarampion	0.88	1.00	0.93	7
Varicela	0.00	0.00	0.00	0
accuracy			0.91	11
macro avg	0.57	0.53	0.55	11
weighted avg	0.92	0.91	0.90	11

--- Results for Group: middle eastern ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	0.00	0.00	0.00	1
Varicela	0.00	0.00	0.00	1
accuracy			0.60	5
macro avg	0.20	0.20	0.20	5
weighted avg	0.60	0.60	0.60	5

--- Results for Group: white ---

	precision	recall	f1-score	support
Herpes	0.60	1.00	0.75	3
Melanoma	1.00	0.67	0.80	3
Monkeypox	1.00	0.50	0.67	4
Sarampion	0.83	0.88	0.86	17
Varicela	0.90	0.90	0.90	21
accuracy			0.85	48
macro avg	0.87	0.79	0.80	48
weighted avg	0.87	0.85	0.85	48

```
=====
FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.0000 EOD (max TPR gap): 1.0000
Worst pair: 0-17|female|white vs 18-59|female|indian
=====
```

```
=====
FAIRNESS SUMMARY (FILTERED, n >= 15) | WGA: 0.8333 EOD (max TPR gap): 0.0784
Worst pair (among filtered): 18-59|female|white vs 18-59|male|white
Groups used (filtered): ['18-59|female|white', '18-59|male|white']
=====
```

```
#####
####
### FOLD 3 ### Train=399, Val=85, Test=81
#####
```

####

Params: 24.73M | Trainable: 1.22M (4.93%)

Approx MACs: 4.13 G, Params (THOP): 24.73 M

=====

START TRAINING (Multimodal ResNet-50) for 200 epochs

=====

Epoch 1/200	Train 1.7273/0.2180	Val 1.6023/0.3529	LR 2.00e-05
Epoch 2/200	Train 1.6629/0.2581	Val 1.5866/0.4471	LR 3.00e-05
Epoch 3/200	Train 1.5995/0.3183	Val 1.5597/0.5529	LR 4.00e-05
Epoch 4/200	Train 1.5399/0.3709	Val 1.5155/0.6824	LR 5.00e-05
Epoch 5/200	Train 1.4301/0.4687	Val 1.4492/0.6941	LR 6.00e-05
Epoch 6/200	Train 1.3094/0.5789	Val 1.3609/0.7412	LR 7.00e-05
Epoch 7/200	Train 1.2295/0.6065	Val 1.2543/0.7765	LR 8.00e-05
Epoch 8/200	Train 1.1140/0.6792	Val 1.1441/0.7882	LR 9.00e-05
Epoch 9/200	Train 1.0669/0.6892	Val 1.0397/0.7882	LR 1.00e-04
Epoch 10/200	Train 0.9987/0.7243	Val 0.9496/0.7882	LR 1.00e-04
Epoch 11/200	Train 0.9583/0.7444	Val 0.8937/0.8000	LR 1.00e-04
Epoch 12/200	Train 0.8876/0.7744	Val 0.8535/0.8000	LR 1.00e-04
Epoch 13/200	Train 0.8524/0.8070	Val 0.8276/0.8000	LR 9.99e-05
Epoch 14/200	Train 0.8119/0.8321	Val 0.8141/0.8000	LR 9.99e-05
Epoch 15/200	Train 0.7767/0.8546	Val 0.7982/0.7882	LR 9.98e-05
Epoch 16/200	Train 0.7716/0.8596	Val 0.7884/0.8118	LR 9.98e-05
Epoch 17/200	Train 0.7478/0.8622	Val 0.7770/0.8235	LR 9.97e-05
Epoch 18/200	Train 0.7457/0.8697	Val 0.7713/0.8353	LR 9.96e-05
Epoch 19/200	Train 0.7155/0.8596	Val 0.7618/0.8471	LR 9.94e-05
Epoch 20/200	Train 0.7122/0.8797	Val 0.7575/0.8471	LR 9.93e-05
Epoch 21/200	Train 0.6847/0.8972	Val 0.7557/0.8471	LR 9.92e-05
Epoch 22/200	Train 0.6749/0.8972	Val 0.7492/0.8471	LR 9.90e-05
Epoch 23/200	Train 0.6743/0.8947	Val 0.7411/0.8471	LR 9.88e-05
Epoch 24/200	Train 0.6355/0.9198	Val 0.7380/0.8471	LR 9.87e-05
Epoch 25/200	Train 0.6510/0.9123	Val 0.7325/0.8471	LR 9.85e-05
Epoch 26/200	Train 0.6208/0.9298	Val 0.7248/0.8471	LR 9.83e-05
Epoch 27/200	Train 0.6183/0.9148	Val 0.7206/0.8471	LR 9.80e-05
Epoch 28/200	Train 0.6126/0.9273	Val 0.7171/0.8471	LR 9.78e-05
Epoch 29/200	Train 0.6001/0.9298	Val 0.7116/0.8588	LR 9.76e-05
Epoch 30/200	Train 0.6051/0.9298	Val 0.7052/0.8588	LR 9.73e-05
Epoch 31/200	Train 0.5966/0.9298	Val 0.6993/0.8706	LR 9.70e-05
Epoch 32/200	Train 0.5806/0.9373	Val 0.6954/0.8706	LR 9.67e-05
Epoch 33/200	Train 0.5781/0.9398	Val 0.6924/0.8706	LR 9.64e-05
Epoch 34/200	Train 0.5928/0.9398	Val 0.6924/0.8706	LR 9.61e-05
Epoch 35/200	Train 0.5831/0.9474	Val 0.6913/0.8706	LR 9.58e-05
Epoch 36/200	Train 0.5786/0.9449	Val 0.6912/0.8706	LR 9.55e-05
Epoch 37/200	Train 0.5927/0.9248	Val 0.6925/0.8588	LR 9.51e-05
Epoch 38/200	Train 0.5580/0.9524	Val 0.6882/0.8706	LR 9.47e-05
Epoch 39/200	Train 0.5571/0.9549	Val 0.6844/0.8588	LR 9.44e-05
Epoch 40/200	Train 0.5495/0.9449	Val 0.6807/0.8588	LR 9.40e-05
Epoch 41/200	Train 0.5646/0.9524	Val 0.6783/0.8588	LR 9.36e-05
Epoch 42/200	Train 0.5470/0.9474	Val 0.6757/0.8588	LR 9.32e-05
Epoch 43/200	Train 0.5447/0.9524	Val 0.6755/0.8588	LR 9.27e-05
Epoch 44/200	Train 0.5419/0.9549	Val 0.6732/0.8588	LR 9.23e-05
Epoch 45/200	Train 0.5450/0.9574	Val 0.6747/0.8471	LR 9.19e-05
Epoch 46/200	Train 0.5441/0.9549	Val 0.6746/0.8471	LR 9.14e-05
Epoch 47/200	Train 0.5306/0.9624	Val 0.6736/0.8588	LR 9.09e-05
Epoch 48/200	Train 0.5199/0.9699	Val 0.6747/0.8588	LR 9.05e-05
Epoch 49/200	Train 0.5347/0.9574	Val 0.6751/0.8588	LR 9.00e-05

Epoch 50/200	Train 0.5434/0.9474	Val 0.6785/0.8588	LR 8.95e-05
Epoch 51/200	Train 0.5349/0.9524	Val 0.6797/0.8471	LR 8.89e-05
Epoch 52/200	Train 0.5087/0.9724	Val 0.6805/0.8471	LR 8.84e-05
Epoch 53/200	Train 0.5140/0.9649	Val 0.6823/0.8353	LR 8.79e-05
Epoch 54/200	Train 0.5262/0.9574	Val 0.6831/0.8353	LR 8.73e-05
Epoch 55/200	Train 0.5013/0.9799	Val 0.6827/0.8588	LR 8.68e-05
Epoch 56/200	Train 0.5016/0.9799	Val 0.6800/0.8588	LR 8.62e-05
Epoch 57/200	Train 0.5209/0.9674	Val 0.6793/0.8588	LR 8.56e-05
Epoch 58/200	Train 0.5061/0.9699	Val 0.6789/0.8588	LR 8.51e-05
Epoch 59/200	Train 0.5043/0.9749	Val 0.6741/0.8824	LR 8.45e-05
Epoch 60/200	Train 0.5070/0.9674	Val 0.6718/0.8941	LR 8.39e-05
Epoch 61/200	Train 0.5059/0.9699	Val 0.6728/0.8941	LR 8.33e-05
Epoch 62/200	Train 0.4964/0.9825	Val 0.6719/0.8941	LR 8.26e-05
Epoch 63/200	Train 0.5006/0.9774	Val 0.6730/0.8941	LR 8.20e-05
Epoch 64/200	Train 0.5054/0.9749	Val 0.6710/0.8941	LR 8.14e-05
Epoch 65/200	Train 0.4936/0.9699	Val 0.6708/0.8941	LR 8.07e-05
Epoch 66/200	Train 0.4940/0.9825	Val 0.6683/0.8941	LR 8.01e-05
Epoch 67/200	Train 0.4932/0.9724	Val 0.6682/0.8824	LR 7.94e-05
Epoch 68/200	Train 0.4976/0.9674	Val 0.6658/0.8824	LR 7.87e-05
Epoch 69/200	Train 0.4993/0.9799	Val 0.6682/0.8706	LR 7.80e-05
Epoch 70/200	Train 0.4895/0.9774	Val 0.6707/0.8824	LR 7.73e-05
Epoch 71/200	Train 0.4814/0.9799	Val 0.6699/0.8824	LR 7.67e-05
Epoch 72/200	Train 0.4927/0.9699	Val 0.6671/0.8588	LR 7.59e-05
Epoch 73/200	Train 0.4899/0.9774	Val 0.6654/0.8471	LR 7.52e-05
Epoch 74/200	Train 0.5012/0.9724	Val 0.6651/0.8588	LR 7.45e-05
Epoch 75/200	Train 0.4833/0.9825	Val 0.6650/0.8588	LR 7.38e-05
Epoch 76/200	Train 0.4924/0.9774	Val 0.6644/0.8588	LR 7.31e-05
Epoch 77/200	Train 0.4904/0.9699	Val 0.6610/0.8588	LR 7.23e-05
Epoch 78/200	Train 0.4791/0.9774	Val 0.6617/0.8588	LR 7.16e-05
Epoch 79/200	Train 0.4957/0.9825	Val 0.6619/0.8471	LR 7.08e-05
Epoch 80/200	Train 0.4760/0.9825	Val 0.6633/0.8471	LR 7.01e-05
Epoch 81/200	Train 0.4845/0.9825	Val 0.6647/0.8471	LR 6.93e-05
Epoch 82/200	Train 0.4837/0.9724	Val 0.6651/0.8471	LR 6.86e-05
Epoch 83/200	Train 0.4807/0.9850	Val 0.6646/0.8471	LR 6.78e-05
Epoch 84/200	Train 0.4836/0.9799	Val 0.6669/0.8471	LR 6.70e-05
Epoch 85/200	Train 0.4815/0.9724	Val 0.6685/0.8471	LR 6.62e-05
Epoch 86/200	Train 0.4728/0.9799	Val 0.6683/0.8588	LR 6.55e-05
Epoch 87/200	Train 0.4760/0.9799	Val 0.6697/0.8824	LR 6.47e-05
Epoch 88/200	Train 0.4795/0.9850	Val 0.6677/0.8471	LR 6.39e-05
Epoch 89/200	Train 0.4761/0.9875	Val 0.6685/0.8471	LR 6.31e-05
Epoch 90/200	Train 0.4748/0.9825	Val 0.6685/0.8706	LR 6.23e-05
Epoch 91/200	Train 0.4766/0.9900	Val 0.6684/0.8706	LR 6.15e-05
Epoch 92/200	Train 0.4685/0.9799	Val 0.6693/0.8588	LR 6.07e-05
Epoch 93/200	Train 0.4681/0.9825	Val 0.6680/0.8706	LR 5.99e-05
Epoch 94/200	Train 0.4766/0.9825	Val 0.6714/0.8941	LR 5.90e-05
Epoch 95/200	Train 0.4679/0.9850	Val 0.6711/0.8941	LR 5.82e-05
Epoch 96/200	Train 0.4796/0.9825	Val 0.6722/0.8706	LR 5.74e-05
Epoch 97/200	Train 0.4757/0.9850	Val 0.6687/0.8706	LR 5.66e-05

Early stopping.

Training complete in 5.6 minutes. Best Val Loss 0.6610

OVERALL PERFORMANCE (ON COMBINED DATASET)

precision recall f1-score support

Herpes	0.67	0.29	0.40	7
Melanoma	1.00	1.00	1.00	3
Monkeypox	0.94	0.94	0.94	18
Sarampion	0.87	0.90	0.88	29
Varicela	0.85	0.96	0.90	24
accuracy			0.88	81
macro avg	0.87	0.82	0.83	81
weighted avg	0.87	0.88	0.86	81

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PERFORMANCE ANALYSIS BY: DOMINANT_GENDER

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--- Results for Group: Man ---

	precision	recall	f1-score	support
Herpes	1.00	0.17	0.29	6
Melanoma	1.00	1.00	1.00	3
Monkeypox	0.93	0.93	0.93	15
Sarampion	0.70	1.00	0.82	7
Varicela	0.88	1.00	0.93	14
accuracy			0.87	45
macro avg	0.90	0.82	0.80	45
weighted avg	0.89	0.87	0.83	45

--- Results for Group: Woman ---

	precision	recall	f1-score	support
Herpes	0.50	1.00	0.67	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	0.95	0.86	0.90	22
Varicela	0.82	0.90	0.86	10
accuracy			0.89	36
macro avg	0.65	0.75	0.69	36
weighted avg	0.91	0.89	0.89	36

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PERFORMANCE ANALYSIS BY: AGE_GROUP

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--- Results for Group: 0-17 ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	1.00	0.33	0.50	3

Varicela	0.83	1.00	0.91	5
accuracy			0.75	8
macro avg	0.37	0.27	0.28	8
weighted avg	0.90	0.75	0.76	8

--- Results for Group: 18-59 ---

	precision	recall	f1-score	support
Herpes	1.00	0.29	0.44	7
Melanoma	1.00	1.00	1.00	3
Monkeypox	0.94	0.94	0.94	18
Sarampion	0.86	0.96	0.91	26
Varicela	0.86	0.95	0.90	19
accuracy			0.89	73
macro avg	0.93	0.83	0.84	73
weighted avg	0.90	0.89	0.87	73

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PERFORMANCE ANALYSIS BY: DOMINANT_RACE

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--- Results for Group: asian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	2
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.75	1.00	0.86	3
Sarampion	1.00	1.00	1.00	4
Varicela	0.67	1.00	0.80	2
accuracy			0.82	11
macro avg	0.48	0.60	0.53	11
weighted avg	0.69	0.82	0.74	11

--- Results for Group: black ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	0.00	0.00	0.00	0
Varicela	0.00	0.00	0.00	0
accuracy			1.00	3
macro avg	0.20	0.20	0.20	3
weighted avg	1.00	1.00	1.00	3

--- Results for Group: indian ---

	precision	recall	f1-score	support
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Herpes	0.00	0.00	0.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	4
Sarampion	0.00	0.00	0.00	1
Varicela	0.00	0.00	0.00	0
accuracy			0.67	6
macro avg	0.20	0.20	0.20	6
weighted avg	0.67	0.67	0.67	6

--- Results for Group: latino hispanic ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	1.00	1.00	1.00	1
Monkeypox	1.00	1.00	1.00	2
Sarampion	1.00	1.00	1.00	7
Varicela	0.00	0.00	0.00	0
accuracy			1.00	10
macro avg	0.60	0.60	0.60	10
weighted avg	1.00	1.00	1.00	10

--- Results for Group: middle eastern ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	1.00	1.00	1.00	1
Monkeypox	1.00	0.50	0.67	2
Sarampion	0.50	1.00	0.67	1
Varicela	0.00	0.00	0.00	0
accuracy			0.75	4
macro avg	0.50	0.50	0.47	4
weighted avg	0.88	0.75	0.75	4

--- Results for Group: white ---

	precision	recall	f1-score	support
Herpes	1.00	0.50	0.67	4
Melanoma	1.00	1.00	1.00	1
Monkeypox	1.00	1.00	1.00	4
Sarampion	0.88	0.88	0.88	16
Varicela	0.88	0.95	0.91	22
accuracy			0.89	47
macro avg	0.95	0.87	0.89	47
weighted avg	0.90	0.89	0.89	47

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FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.0000 EOD (max TPR gap): 1.0000
Worst pair: 0-17|female|indian vs 0-17|male|white

FAIRNESS SUMMARY (FILTERED, n >= 15) | WGA: 0.9000 EOD (max TPR gap): 0.3034
Worst pair (among filtered): 18-59|female|white vs 18-59|male|white
Groups used (filtered): ['18-59|female|white', '18-59|male|white']

FOLD 4 ### Train=399, Val=85, Test=81

Params: 24.73M | Trainable: 1.22M (4.93%)
Approx MACs: 4.13 G, Params (THOP): 24.73 M

START TRAINING (Multimodal ResNet-50) for 200 epochs

Epoch 1/200	Train 1.6784/0.2180	Val 1.6036/0.2588	LR 2.00e-05
Epoch 2/200	Train 1.6325/0.2531	Val 1.5891/0.3765	LR 3.00e-05
Epoch 3/200	Train 1.5544/0.3158	Val 1.5619/0.3882	LR 4.00e-05
Epoch 4/200	Train 1.4567/0.4160	Val 1.5194/0.5529	LR 5.00e-05
Epoch 5/200	Train 1.3324/0.5639	Val 1.4531/0.6941	LR 6.00e-05
Epoch 6/200	Train 1.2133/0.6241	Val 1.3623/0.7412	LR 7.00e-05
Epoch 7/200	Train 1.1267/0.6992	Val 1.2540/0.7529	LR 8.00e-05
Epoch 8/200	Train 1.0431/0.7193	Val 1.1382/0.8000	LR 9.00e-05
Epoch 9/200	Train 0.9580/0.7569	Val 1.0349/0.8353	LR 1.00e-04
Epoch 10/200	Train 0.9079/0.7820	Val 0.9548/0.8471	LR 1.00e-04
Epoch 11/200	Train 0.8594/0.8045	Val 0.8926/0.8353	LR 1.00e-04
Epoch 12/200	Train 0.8379/0.8070	Val 0.8441/0.8471	LR 1.00e-04
Epoch 13/200	Train 0.7887/0.8321	Val 0.8068/0.8824	LR 9.99e-05
Epoch 14/200	Train 0.7632/0.8747	Val 0.7818/0.8706	LR 9.99e-05
Epoch 15/200	Train 0.7344/0.8772	Val 0.7630/0.8824	LR 9.98e-05
Epoch 16/200	Train 0.7237/0.8797	Val 0.7524/0.8824	LR 9.98e-05
Epoch 17/200	Train 0.7049/0.8847	Val 0.7413/0.8824	LR 9.97e-05
Epoch 18/200	Train 0.6743/0.8947	Val 0.7322/0.8941	LR 9.96e-05
Epoch 19/200	Train 0.6511/0.9023	Val 0.7278/0.8706	LR 9.94e-05
Epoch 20/200	Train 0.6411/0.9073	Val 0.7207/0.8706	LR 9.93e-05
Epoch 21/200	Train 0.6549/0.8972	Val 0.7154/0.8824	LR 9.92e-05
Epoch 22/200	Train 0.6241/0.9098	Val 0.7128/0.8824	LR 9.90e-05
Epoch 23/200	Train 0.6183/0.9273	Val 0.7088/0.8706	LR 9.88e-05
Epoch 24/200	Train 0.6150/0.9248	Val 0.7060/0.8706	LR 9.87e-05
Epoch 25/200	Train 0.6082/0.9298	Val 0.7047/0.8706	LR 9.85e-05
Epoch 26/200	Train 0.5979/0.9298	Val 0.7000/0.8706	LR 9.83e-05
Epoch 27/200	Train 0.5831/0.9398	Val 0.6975/0.8706	LR 9.80e-05
Epoch 28/200	Train 0.5846/0.9474	Val 0.6942/0.8706	LR 9.78e-05
Epoch 29/200	Train 0.5818/0.9223	Val 0.6914/0.8706	LR 9.76e-05
Epoch 30/200	Train 0.5699/0.9348	Val 0.6889/0.8824	LR 9.73e-05
Epoch 31/200	Train 0.5630/0.9474	Val 0.6896/0.8824	LR 9.70e-05
Epoch 32/200	Train 0.5519/0.9474	Val 0.6867/0.8824	LR 9.67e-05
Epoch 33/200	Train 0.5618/0.9424	Val 0.6843/0.8706	LR 9.64e-05
Epoch 34/200	Train 0.5438/0.9574	Val 0.6803/0.8824	LR 9.61e-05
Epoch 35/200	Train 0.5446/0.9499	Val 0.6800/0.8824	LR 9.58e-05

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Epoch 36/200 | Train 0.5361/0.9674 | Val 0.6796/0.8706 | LR 9.55e-05
Epoch 37/200 | Train 0.5416/0.9649 | Val 0.6753/0.8706 | LR 9.51e-05
Epoch 38/200 | Train 0.5355/0.9549 | Val 0.6724/0.8706 | LR 9.47e-05
Epoch 39/200 | Train 0.5372/0.9574 | Val 0.6712/0.8706 | LR 9.44e-05
Epoch 40/200 | Train 0.5357/0.9549 | Val 0.6702/0.8824 | LR 9.40e-05
Epoch 41/200 | Train 0.5048/0.9774 | Val 0.6692/0.8824 | LR 9.36e-05
Epoch 42/200 | Train 0.5223/0.9624 | Val 0.6692/0.8824 | LR 9.32e-05
Epoch 43/200 | Train 0.5393/0.9599 | Val 0.6717/0.8824 | LR 9.27e-05
Epoch 44/200 | Train 0.5242/0.9624 | Val 0.6745/0.8706 | LR 9.23e-05
Epoch 45/200 | Train 0.5142/0.9749 | Val 0.6769/0.8706 | LR 9.19e-05
Epoch 46/200 | Train 0.5208/0.9749 | Val 0.6764/0.8824 | LR 9.14e-05
Epoch 47/200 | Train 0.5033/0.9825 | Val 0.6760/0.8706 | LR 9.09e-05
Epoch 48/200 | Train 0.5113/0.9749 | Val 0.6740/0.8706 | LR 9.05e-05
Epoch 49/200 | Train 0.5134/0.9699 | Val 0.6716/0.8706 | LR 9.00e-05
Epoch 50/200 | Train 0.4921/0.9825 | Val 0.6690/0.8706 | LR 8.95e-05
Epoch 51/200 | Train 0.5011/0.9774 | Val 0.6672/0.8824 | LR 8.89e-05
Epoch 52/200 | Train 0.5097/0.9649 | Val 0.6651/0.8824 | LR 8.84e-05
Epoch 53/200 | Train 0.4972/0.9724 | Val 0.6642/0.8941 | LR 8.79e-05
Epoch 54/200 | Train 0.5060/0.9724 | Val 0.6658/0.8941 | LR 8.73e-05
Epoch 55/200 | Train 0.4932/0.9749 | Val 0.6662/0.8941 | LR 8.68e-05
Epoch 56/200 | Train 0.4930/0.9774 | Val 0.6666/0.8941 | LR 8.62e-05
Epoch 57/200 | Train 0.4902/0.9774 | Val 0.6700/0.8941 | LR 8.56e-05
Epoch 58/200 | Train 0.5108/0.9524 | Val 0.6751/0.8941 | LR 8.51e-05
Epoch 59/200 | Train 0.4871/0.9774 | Val 0.6745/0.8824 | LR 8.45e-05
Epoch 60/200 | Train 0.4918/0.9825 | Val 0.6765/0.8824 | LR 8.39e-05
Epoch 61/200 | Train 0.4893/0.9825 | Val 0.6813/0.8706 | LR 8.33e-05
Epoch 62/200 | Train 0.4820/0.9724 | Val 0.6866/0.8706 | LR 8.26e-05
Epoch 63/200 | Train 0.4861/0.9875 | Val 0.6844/0.8706 | LR 8.20e-05
Epoch 64/200 | Train 0.4816/0.9774 | Val 0.6816/0.8588 | LR 8.14e-05
Epoch 65/200 | Train 0.4928/0.9774 | Val 0.6833/0.8706 | LR 8.07e-05
Epoch 66/200 | Train 0.4824/0.9825 | Val 0.6839/0.8588 | LR 8.01e-05
Epoch 67/200 | Train 0.4761/0.9774 | Val 0.6799/0.8706 | LR 7.94e-05
Epoch 68/200 | Train 0.4784/0.9850 | Val 0.6786/0.8706 | LR 7.87e-05
Epoch 69/200 | Train 0.4740/0.9850 | Val 0.6785/0.8706 | LR 7.80e-05
Epoch 70/200 | Train 0.4827/0.9799 | Val 0.6813/0.8706 | LR 7.73e-05
Epoch 71/200 | Train 0.4812/0.9825 | Val 0.6787/0.8706 | LR 7.67e-05
Epoch 72/200 | Train 0.4836/0.9749 | Val 0.6756/0.8824 | LR 7.59e-05
Epoch 73/200 | Train 0.4642/0.9950 | Val 0.6757/0.8824 | LR 7.52e-05

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Early stopping.

Training complete in 4.2 minutes. Best Val Loss 0.6642

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OVERALL PERFORMANCE (ON COMBINED DATASET)

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	precision	recall	f1-score	support
Herpes	0.71	0.83	0.77	6
Melanoma	1.00	0.50	0.67	2
Monkeypox	0.94	0.79	0.86	19
Sarampion	0.88	0.93	0.90	30
Varicela	0.76	0.79	0.78	24
accuracy			0.84	81
macro avg	0.86	0.77	0.79	81
weighted avg	0.85	0.84	0.84	81

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PERFORMANCE ANALYSIS BY: DOMINANT_GENDER
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--- Results for Group: Man ---
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	precision	recall	f1-score	support
Herpes	0.67	0.80	0.73	5
Melanoma	1.00	0.50	0.67	2
Monkeypox	0.93	0.81	0.87	16
Sarampion	0.93	0.88	0.90	16
Varicela	0.69	0.85	0.76	13
accuracy			0.83	52
macro avg	0.84	0.77	0.78	52
weighted avg	0.85	0.83	0.83	52

```
--- Results for Group: Woman ---
```

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	0.67	0.80	3
Sarampion	0.82	1.00	0.90	14
Varicela	0.89	0.73	0.80	11
accuracy			0.86	29
macro avg	0.74	0.68	0.70	29
weighted avg	0.87	0.86	0.86	29

```
=====
PERFORMANCE ANALYSIS BY: AGE_GROUP
=====
```

```
--- Results for Group: 0-17 ---
```

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	1.00	1.00	1.00	2
Varicela	1.00	1.00	1.00	3
accuracy			1.00	5
macro avg	0.40	0.40	0.40	5
weighted avg	1.00	1.00	1.00	5

```
--- Results for Group: 18-59 ---
```

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

Herpes	0.71	0.83	0.77	6
Melanoma	1.00	0.50	0.67	2
Monkeypox	0.94	0.79	0.86	19
Sarampion	0.87	0.93	0.90	28
Varicela	0.73	0.76	0.74	21
accuracy			0.83	76
macro avg	0.85	0.76	0.79	76
weighted avg	0.84	0.83	0.83	76

=====

PERFORMANCE ANALYSIS BY: DOMINANT_RACE

=====

--- Results for Group: asian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	0.60	1.00	0.75	3
Varicela	0.75	0.60	0.67	5
accuracy			0.70	10
macro avg	0.47	0.52	0.48	10
weighted avg	0.66	0.70	0.66	10

--- Results for Group: black ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	1.00	1.00	1.00	2
Varicela	0.00	0.00	0.00	0
accuracy			1.00	5
macro avg	0.40	0.40	0.40	5
weighted avg	1.00	1.00	1.00	5

--- Results for Group: indian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	0.80	1.00	0.89	4
Varicela	0.00	0.00	0.00	1
accuracy			0.86	7
macro avg	0.36	0.40	0.38	7
weighted avg	0.74	0.86	0.79	7

```

--- Results for Group: latino hispanic ---
      precision    recall  f1-score   support

    Herpes         0.00      0.00      0.00         0
    Melanoma        0.00      0.00      0.00         0
    Monkeypox        1.00      0.67      0.80         3
    Sarampion        1.00      1.00      1.00         4
    Varicela        0.00      0.00      0.00         0

 accuracy                   0.86         7
 macro avg         0.40      0.33      0.36         7
 weighted avg      1.00      0.86      0.91         7

```

```

--- Results for Group: middle eastern ---
      precision    recall  f1-score   support

    Herpes         0.00      0.00      0.00         0
    Melanoma        1.00      1.00      1.00         1
    Monkeypox        1.00      1.00      1.00         2
    Sarampion        1.00      1.00      1.00         1
    Varicela        0.00      0.00      0.00         0

 accuracy                   1.00         4
 macro avg         0.60      0.60      0.60         4
 weighted avg      1.00      1.00      1.00         4

```

```

--- Results for Group: white ---
      precision    recall  f1-score   support

    Herpes         0.71      1.00      0.83         5
    Melanoma        0.00      0.00      0.00         1
    Monkeypox        0.83      0.62      0.71         8
    Sarampion        0.93      0.88      0.90        16
    Varicela        0.80      0.89      0.84        18

 accuracy                   0.83        48
 macro avg         0.66      0.68      0.66        48
 weighted avg      0.82      0.83      0.82        48

```

```

=====
FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.5000 EOD (max TPR gap): 1.0000
Worst pair: 0-17|female|white vs 18-59|female|indian
=====

```

```

=====
FAIRNESS SUMMARY (FILTERED, n >= 15) | WGA: 0.7308 EOD (max TPR gap): 0.1852
Worst pair (among filtered): 18-59|female|white vs 18-59|male|white
Groups used (filtered): ['18-59|female|white', '18-59|male|white']
=====

```

```
#####  
####  
### FOLD 5 ### Train=399, Val=85, Test=81  
#####  
####  
Params: 24.73M | Trainable: 1.22M (4.93%)  
Approx MACs: 4.13 G, Params (THOP): 24.73 M
```

```
=====
```

START TRAINING (Multimodal ResNet-50) for 200 epochs

```
=====
```

Epoch 1/200	Train 1.6533/0.2581	Val 1.6000/0.3176	LR 2.00e-05
Epoch 2/200	Train 1.5885/0.2957	Val 1.5882/0.4353	LR 3.00e-05
Epoch 3/200	Train 1.5611/0.3709	Val 1.5638/0.5412	LR 4.00e-05
Epoch 4/200	Train 1.4348/0.4812	Val 1.5234/0.5765	LR 5.00e-05
Epoch 5/200	Train 1.3410/0.5439	Val 1.4587/0.6706	LR 6.00e-05
Epoch 6/200	Train 1.2426/0.6316	Val 1.3710/0.7059	LR 7.00e-05
Epoch 7/200	Train 1.1393/0.6541	Val 1.2654/0.7294	LR 8.00e-05
Epoch 8/200	Train 1.0678/0.7068	Val 1.1544/0.7412	LR 9.00e-05
Epoch 9/200	Train 0.9836/0.7544	Val 1.0517/0.7647	LR 1.00e-04
Epoch 10/200	Train 0.9444/0.7719	Val 0.9646/0.7765	LR 1.00e-04
Epoch 11/200	Train 0.9016/0.7870	Val 0.9008/0.7882	LR 1.00e-04
Epoch 12/200	Train 0.8448/0.8195	Val 0.8625/0.8000	LR 1.00e-04
Epoch 13/200	Train 0.8365/0.8045	Val 0.8361/0.8000	LR 9.99e-05
Epoch 14/200	Train 0.7994/0.8195	Val 0.8182/0.8118	LR 9.99e-05
Epoch 15/200	Train 0.7893/0.8346	Val 0.7981/0.8235	LR 9.98e-05
Epoch 16/200	Train 0.7585/0.8371	Val 0.7807/0.8353	LR 9.98e-05
Epoch 17/200	Train 0.7432/0.8672	Val 0.7691/0.8471	LR 9.97e-05
Epoch 18/200	Train 0.7413/0.8647	Val 0.7551/0.8353	LR 9.96e-05
Epoch 19/200	Train 0.7064/0.8822	Val 0.7442/0.8353	LR 9.94e-05
Epoch 20/200	Train 0.6814/0.8822	Val 0.7404/0.8353	LR 9.93e-05
Epoch 21/200	Train 0.6673/0.8922	Val 0.7348/0.8353	LR 9.92e-05
Epoch 22/200	Train 0.6751/0.8872	Val 0.7278/0.8353	LR 9.90e-05
Epoch 23/200	Train 0.6552/0.8947	Val 0.7242/0.8353	LR 9.88e-05
Epoch 24/200	Train 0.6448/0.9173	Val 0.7202/0.8353	LR 9.87e-05
Epoch 25/200	Train 0.6320/0.9173	Val 0.7140/0.8353	LR 9.85e-05
Epoch 26/200	Train 0.6307/0.9173	Val 0.7077/0.8353	LR 9.83e-05
Epoch 27/200	Train 0.6147/0.9248	Val 0.7017/0.8353	LR 9.80e-05
Epoch 28/200	Train 0.6019/0.9424	Val 0.6973/0.8353	LR 9.78e-05
Epoch 29/200	Train 0.5926/0.9348	Val 0.6971/0.8353	LR 9.76e-05
Epoch 30/200	Train 0.6058/0.9348	Val 0.6896/0.8353	LR 9.73e-05
Epoch 31/200	Train 0.5888/0.9449	Val 0.6903/0.8588	LR 9.70e-05
Epoch 32/200	Train 0.5691/0.9474	Val 0.6873/0.8588	LR 9.67e-05
Epoch 33/200	Train 0.5662/0.9424	Val 0.6857/0.8588	LR 9.64e-05
Epoch 34/200	Train 0.5734/0.9398	Val 0.6851/0.8588	LR 9.61e-05
Epoch 35/200	Train 0.5695/0.9499	Val 0.6819/0.8588	LR 9.58e-05
Epoch 36/200	Train 0.5519/0.9524	Val 0.6808/0.8471	LR 9.55e-05
Epoch 37/200	Train 0.5571/0.9398	Val 0.6737/0.8471	LR 9.51e-05
Epoch 38/200	Train 0.5394/0.9524	Val 0.6735/0.8471	LR 9.47e-05
Epoch 39/200	Train 0.5403/0.9624	Val 0.6722/0.8471	LR 9.44e-05
Epoch 40/200	Train 0.5398/0.9649	Val 0.6704/0.8471	LR 9.40e-05
Epoch 41/200	Train 0.5418/0.9599	Val 0.6674/0.8353	LR 9.36e-05
Epoch 42/200	Train 0.5520/0.9474	Val 0.6617/0.8588	LR 9.32e-05
Epoch 43/200	Train 0.5494/0.9549	Val 0.6630/0.8471	LR 9.27e-05
Epoch 44/200	Train 0.5306/0.9699	Val 0.6634/0.8471	LR 9.23e-05
Epoch 45/200	Train 0.5231/0.9749	Val 0.6632/0.8588	LR 9.19e-05

Epoch 46/200	Train 0.5313/0.9599	Val 0.6612/0.8588	LR 9.14e-05
Epoch 47/200	Train 0.5268/0.9674	Val 0.6608/0.8588	LR 9.09e-05
Epoch 48/200	Train 0.5309/0.9674	Val 0.6610/0.8471	LR 9.05e-05
Epoch 49/200	Train 0.5125/0.9674	Val 0.6607/0.8588	LR 9.00e-05
Epoch 50/200	Train 0.5236/0.9674	Val 0.6622/0.8471	LR 8.95e-05
Epoch 51/200	Train 0.5240/0.9649	Val 0.6618/0.8471	LR 8.89e-05
Epoch 52/200	Train 0.5107/0.9724	Val 0.6603/0.8588	LR 8.84e-05
Epoch 53/200	Train 0.5121/0.9624	Val 0.6588/0.8471	LR 8.79e-05
Epoch 54/200	Train 0.4976/0.9799	Val 0.6554/0.8588	LR 8.73e-05
Epoch 55/200	Train 0.5022/0.9825	Val 0.6550/0.8588	LR 8.68e-05
Epoch 56/200	Train 0.5143/0.9774	Val 0.6576/0.8471	LR 8.62e-05
Epoch 57/200	Train 0.4911/0.9850	Val 0.6549/0.8588	LR 8.56e-05
Epoch 58/200	Train 0.4937/0.9875	Val 0.6546/0.8471	LR 8.51e-05
Epoch 59/200	Train 0.4998/0.9774	Val 0.6555/0.8471	LR 8.45e-05
Epoch 60/200	Train 0.5049/0.9674	Val 0.6526/0.8706	LR 8.39e-05
Epoch 61/200	Train 0.4939/0.9850	Val 0.6501/0.8588	LR 8.33e-05
Epoch 62/200	Train 0.4877/0.9825	Val 0.6486/0.8588	LR 8.26e-05
Epoch 63/200	Train 0.4925/0.9724	Val 0.6452/0.8471	LR 8.20e-05
Epoch 64/200	Train 0.5057/0.9674	Val 0.6441/0.8588	LR 8.14e-05
Epoch 65/200	Train 0.4963/0.9825	Val 0.6459/0.8588	LR 8.07e-05
Epoch 66/200	Train 0.4993/0.9749	Val 0.6469/0.8706	LR 8.01e-05
Epoch 67/200	Train 0.4881/0.9875	Val 0.6476/0.8588	LR 7.94e-05
Epoch 68/200	Train 0.4876/0.9799	Val 0.6468/0.8588	LR 7.87e-05
Epoch 69/200	Train 0.4912/0.9749	Val 0.6458/0.8706	LR 7.80e-05
Epoch 70/200	Train 0.4764/0.9850	Val 0.6405/0.8824	LR 7.73e-05
Epoch 71/200	Train 0.4744/0.9900	Val 0.6379/0.8706	LR 7.67e-05
Epoch 72/200	Train 0.4835/0.9875	Val 0.6394/0.8824	LR 7.59e-05
Epoch 73/200	Train 0.4843/0.9850	Val 0.6364/0.8824	LR 7.52e-05
Epoch 74/200	Train 0.4960/0.9825	Val 0.6358/0.8824	LR 7.45e-05
Epoch 75/200	Train 0.4861/0.9799	Val 0.6348/0.8706	LR 7.38e-05
Epoch 76/200	Train 0.4815/0.9875	Val 0.6328/0.8706	LR 7.31e-05
Epoch 77/200	Train 0.4751/0.9825	Val 0.6356/0.8824	LR 7.23e-05
Epoch 78/200	Train 0.4810/0.9875	Val 0.6348/0.8824	LR 7.16e-05
Epoch 79/200	Train 0.4853/0.9825	Val 0.6354/0.8706	LR 7.08e-05
Epoch 80/200	Train 0.4689/0.9825	Val 0.6377/0.8824	LR 7.01e-05
Epoch 81/200	Train 0.4801/0.9799	Val 0.6372/0.8824	LR 6.93e-05
Epoch 82/200	Train 0.4702/0.9875	Val 0.6358/0.8824	LR 6.86e-05
Epoch 83/200	Train 0.4789/0.9799	Val 0.6374/0.8706	LR 6.78e-05
Epoch 84/200	Train 0.4886/0.9749	Val 0.6385/0.8588	LR 6.70e-05
Epoch 85/200	Train 0.4684/0.9850	Val 0.6413/0.8588	LR 6.62e-05
Epoch 86/200	Train 0.4837/0.9724	Val 0.6424/0.8588	LR 6.55e-05
Epoch 87/200	Train 0.4824/0.9825	Val 0.6444/0.8588	LR 6.47e-05
Epoch 88/200	Train 0.4719/0.9950	Val 0.6455/0.8706	LR 6.39e-05
Epoch 89/200	Train 0.4716/0.9875	Val 0.6437/0.8941	LR 6.31e-05
Epoch 90/200	Train 0.4703/0.9850	Val 0.6451/0.8706	LR 6.23e-05
Epoch 91/200	Train 0.4748/0.9799	Val 0.6438/0.8706	LR 6.15e-05
Epoch 92/200	Train 0.4608/0.9900	Val 0.6430/0.8706	LR 6.07e-05
Epoch 93/200	Train 0.4636/0.9875	Val 0.6439/0.8706	LR 5.99e-05
Epoch 94/200	Train 0.4642/0.9875	Val 0.6447/0.8824	LR 5.90e-05
Epoch 95/200	Train 0.4696/0.9900	Val 0.6474/0.8824	LR 5.82e-05
Epoch 96/200	Train 0.4655/0.9850	Val 0.6476/0.8706	LR 5.74e-05

Early stopping.

Training complete in 5.6 minutes. Best Val Loss 0.6328

=====

OVERALL PERFORMANCE (ON COMBINED DATASET)

```
=====
```

	precision	recall	f1-score	support
Herpes	1.00	0.83	0.91	6
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.95	0.95	0.95	19
Sarampion	0.96	0.90	0.93	30
Varicela	0.89	1.00	0.94	24
accuracy			0.94	81
macro avg	0.96	0.94	0.95	81
weighted avg	0.94	0.94	0.94	81

```
=====
```

PERFORMANCE ANALYSIS BY: DOMINANT_GENDER

```
=====
```

--- Results for Group: Man ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	4
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.93	0.93	0.93	15
Sarampion	1.00	0.87	0.93	15
Varicela	0.89	1.00	0.94	17
accuracy			0.94	53
macro avg	0.97	0.96	0.96	53
weighted avg	0.95	0.94	0.94	53

--- Results for Group: Woman ---

	precision	recall	f1-score	support
Herpes	1.00	0.50	0.67	2
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	4
Sarampion	0.93	0.93	0.93	15
Varicela	0.88	1.00	0.93	7
accuracy			0.93	28
macro avg	0.76	0.69	0.71	28
weighted avg	0.93	0.93	0.92	28

```
=====
```

PERFORMANCE ANALYSIS BY: AGE_GROUP

```
=====
```

--- Results for Group: 0-17 ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0

Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	0.00	0.00	0.00	1
Varicela	1.00	1.00	1.00	5
accuracy			0.83	6
macro avg	0.20	0.20	0.20	6
weighted avg	0.83	0.83	0.83	6

--- Results for Group: 18-59 ---

	precision	recall	f1-score	support
Herpes	1.00	0.83	0.91	6
Melanoma	1.00	1.00	1.00	2
Monkeypox	1.00	0.95	0.97	19
Sarampion	0.96	0.93	0.95	29
Varicela	0.86	1.00	0.93	19
accuracy			0.95	75
macro avg	0.97	0.94	0.95	75
weighted avg	0.95	0.95	0.95	75

=====

PERFORMANCE ANALYSIS BY: DOMINANT_RACE

=====

--- Results for Group: asian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	0.67	0.50	0.57	4
Varicela	0.60	1.00	0.75	3
accuracy			0.67	9
macro avg	0.45	0.50	0.46	9
weighted avg	0.61	0.67	0.62	9

--- Results for Group: black ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.80	1.00	0.89	4
Sarampion	0.00	0.00	0.00	1
Varicela	0.00	0.00	0.00	0
accuracy			0.83	6
macro avg	0.36	0.40	0.38	6
weighted avg	0.70	0.83	0.76	6

--- Results for Group: indian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	1.00	1.00	1.00	4
Varicela	1.00	1.00	1.00	3
accuracy			1.00	10
macro avg	0.60	0.60	0.60	10
weighted avg	1.00	1.00	1.00	10

--- Results for Group: latino hispanic ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	1.00	1.00	1.00	7
Varicela	0.00	0.00	0.00	0
accuracy			1.00	9
macro avg	0.40	0.40	0.40	9
weighted avg	1.00	1.00	1.00	9

--- Results for Group: middle eastern ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	1.00	1.00	1.00	1
Varicela	0.00	0.00	0.00	0
accuracy			1.00	2
macro avg	0.40	0.40	0.40	2
weighted avg	1.00	1.00	1.00	2

--- Results for Group: white ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	4
Melanoma	1.00	1.00	1.00	2
Monkeypox	1.00	0.88	0.93	8
Sarampion	1.00	1.00	1.00	13
Varicela	0.95	1.00	0.97	18
accuracy			0.98	45
macro avg	0.99	0.97	0.98	45
weighted avg	0.98	0.98	0.98	45

```
=====
FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.0000 EOD (max TPR gap): 1.0000
Worst pair: 0-17|male|black vs 18-59|female|latino hispanic
=====
```

```
=====
FAIRNESS SUMMARY (FILTERED, n >= 15) | WGA: 0.9615 EOD (max TPR gap): 0.0000
Groups used (filtered): ['18-59|male|white']
=====
```

```
#####
####
### FOLD 6 ### Train=400, Val=85, Test=80
#####
####
Params: 24.73M | Trainable: 1.22M (4.93%)
Approx MACs: 4.13 G, Params (THOP): 24.73 M
```

```
=====
START TRAINING (Multimodal ResNet-50) for 200 epochs
=====
```

Epoch 1/200	Train 1.7208/0.2075	Val 1.6150/0.2235	LR 2.00e-05
Epoch 2/200	Train 1.7101/0.2100	Val 1.6038/0.3059	LR 3.00e-05
Epoch 3/200	Train 1.5884/0.2725	Val 1.5801/0.3882	LR 4.00e-05
Epoch 4/200	Train 1.5249/0.3625	Val 1.5403/0.5176	LR 5.00e-05
Epoch 5/200	Train 1.4103/0.4775	Val 1.4803/0.6000	LR 6.00e-05
Epoch 6/200	Train 1.2783/0.5850	Val 1.3943/0.6706	LR 7.00e-05
Epoch 7/200	Train 1.1903/0.6450	Val 1.2850/0.7294	LR 8.00e-05
Epoch 8/200	Train 1.0767/0.7050	Val 1.1732/0.7412	LR 9.00e-05
Epoch 9/200	Train 1.0211/0.7225	Val 1.0696/0.7647	LR 1.00e-04
Epoch 10/200	Train 0.9638/0.7375	Val 0.9857/0.7882	LR 1.00e-04
Epoch 11/200	Train 0.9295/0.7700	Val 0.9243/0.7882	LR 1.00e-04
Epoch 12/200	Train 0.8510/0.7900	Val 0.8765/0.7882	LR 1.00e-04
Epoch 13/200	Train 0.8669/0.7900	Val 0.8502/0.7882	LR 9.99e-05
Epoch 14/200	Train 0.7812/0.8325	Val 0.8253/0.8000	LR 9.99e-05
Epoch 15/200	Train 0.7732/0.8375	Val 0.7994/0.8353	LR 9.98e-05
Epoch 16/200	Train 0.7510/0.8575	Val 0.7762/0.8353	LR 9.98e-05
Epoch 17/200	Train 0.7292/0.8650	Val 0.7593/0.8353	LR 9.97e-05
Epoch 18/200	Train 0.7058/0.8800	Val 0.7497/0.8588	LR 9.96e-05
Epoch 19/200	Train 0.6886/0.8950	Val 0.7389/0.8588	LR 9.94e-05
Epoch 20/200	Train 0.6918/0.8950	Val 0.7336/0.8588	LR 9.93e-05
Epoch 21/200	Train 0.6812/0.9000	Val 0.7324/0.8471	LR 9.92e-05
Epoch 22/200	Train 0.6596/0.9025	Val 0.7238/0.8471	LR 9.90e-05
Epoch 23/200	Train 0.6409/0.9150	Val 0.7235/0.8588	LR 9.88e-05
Epoch 24/200	Train 0.6317/0.9200	Val 0.7206/0.8588	LR 9.87e-05
Epoch 25/200	Train 0.6432/0.9100	Val 0.7183/0.8588	LR 9.85e-05
Epoch 26/200	Train 0.6188/0.9400	Val 0.7188/0.8588	LR 9.83e-05
Epoch 27/200	Train 0.6079/0.9325	Val 0.7171/0.8588	LR 9.80e-05
Epoch 28/200	Train 0.6040/0.9400	Val 0.7175/0.8353	LR 9.78e-05
Epoch 29/200	Train 0.5829/0.9425	Val 0.7206/0.8353	LR 9.76e-05
Epoch 30/200	Train 0.5807/0.9450	Val 0.7241/0.8235	LR 9.73e-05
Epoch 31/200	Train 0.5725/0.9525	Val 0.7237/0.8235	LR 9.70e-05
Epoch 32/200	Train 0.5994/0.9300	Val 0.7188/0.8235	LR 9.67e-05
Epoch 33/200	Train 0.5734/0.9475	Val 0.7174/0.8235	LR 9.64e-05

Epoch 34/200	Train 0.5628/0.9525	Val 0.7120/0.8353	LR 9.61e-05
Epoch 35/200	Train 0.5596/0.9550	Val 0.7079/0.8471	LR 9.58e-05
Epoch 36/200	Train 0.5527/0.9625	Val 0.7084/0.8471	LR 9.55e-05
Epoch 37/200	Train 0.5679/0.9550	Val 0.7084/0.8471	LR 9.51e-05
Epoch 38/200	Train 0.5497/0.9500	Val 0.7078/0.8471	LR 9.47e-05
Epoch 39/200	Train 0.5480/0.9550	Val 0.7082/0.8471	LR 9.44e-05
Epoch 40/200	Train 0.5616/0.9625	Val 0.7099/0.8471	LR 9.40e-05
Epoch 41/200	Train 0.5422/0.9550	Val 0.7116/0.8353	LR 9.36e-05
Epoch 42/200	Train 0.5387/0.9675	Val 0.7081/0.8353	LR 9.32e-05
Epoch 43/200	Train 0.5295/0.9625	Val 0.7078/0.8235	LR 9.27e-05
Epoch 44/200	Train 0.5257/0.9650	Val 0.7094/0.8235	LR 9.23e-05
Epoch 45/200	Train 0.5168/0.9550	Val 0.7061/0.8353	LR 9.19e-05
Epoch 46/200	Train 0.5211/0.9725	Val 0.7035/0.8353	LR 9.14e-05
Epoch 47/200	Train 0.5241/0.9675	Val 0.7055/0.8235	LR 9.09e-05
Epoch 48/200	Train 0.5427/0.9575	Val 0.7084/0.8353	LR 9.05e-05
Epoch 49/200	Train 0.5223/0.9625	Val 0.7036/0.8471	LR 9.00e-05
Epoch 50/200	Train 0.5148/0.9725	Val 0.7060/0.8471	LR 8.95e-05
Epoch 51/200	Train 0.5137/0.9750	Val 0.7106/0.8471	LR 8.89e-05
Epoch 52/200	Train 0.5172/0.9725	Val 0.7079/0.8471	LR 8.84e-05
Epoch 53/200	Train 0.5059/0.9750	Val 0.7155/0.8471	LR 8.79e-05
Epoch 54/200	Train 0.5056/0.9650	Val 0.7188/0.8353	LR 8.73e-05
Epoch 55/200	Train 0.5021/0.9725	Val 0.7181/0.8235	LR 8.68e-05
Epoch 56/200	Train 0.5007/0.9725	Val 0.7225/0.8235	LR 8.62e-05
Epoch 57/200	Train 0.4977/0.9800	Val 0.7239/0.8235	LR 8.56e-05
Epoch 58/200	Train 0.4879/0.9850	Val 0.7219/0.8235	LR 8.51e-05
Epoch 59/200	Train 0.4846/0.9850	Val 0.7203/0.8235	LR 8.45e-05
Epoch 60/200	Train 0.5097/0.9750	Val 0.7201/0.8235	LR 8.39e-05
Epoch 61/200	Train 0.4945/0.9825	Val 0.7171/0.8235	LR 8.33e-05
Epoch 62/200	Train 0.4954/0.9825	Val 0.7139/0.8353	LR 8.26e-05
Epoch 63/200	Train 0.4940/0.9800	Val 0.7135/0.8471	LR 8.20e-05
Epoch 64/200	Train 0.4881/0.9850	Val 0.7149/0.8588	LR 8.14e-05
Epoch 65/200	Train 0.4951/0.9725	Val 0.7099/0.8353	LR 8.07e-05
Epoch 66/200	Train 0.4739/0.9875	Val 0.7069/0.8353	LR 8.01e-05

Early stopping.
Training complete in 4.0 minutes. Best Val Loss 0.7035

OVERALL PERFORMANCE (ON COMBINED DATASET)

	precision	recall	f1-score	support
Herpes	1.00	0.67	0.80	6
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.94	0.94	0.94	18
Sarampion	0.96	0.83	0.89	29
Varicela	0.77	0.96	0.86	25
accuracy			0.89	80
macro avg	0.94	0.88	0.90	80
weighted avg	0.90	0.89	0.89	80

PERFORMANCE ANALYSIS BY: DOMINANT_GENDER

--- Results for Group: Man ---

	precision	recall	f1-score	support
Herpes	1.00	0.60	0.75	5
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.94	1.00	0.97	16
Sarampion	1.00	0.79	0.88	19
Varicela	0.69	1.00	0.81	11
accuracy			0.89	53
macro avg	0.93	0.88	0.88	53
weighted avg	0.92	0.89	0.89	53

--- Results for Group: Woman ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	0.50	0.67	2
Sarampion	0.90	0.90	0.90	10
Varicela	0.87	0.93	0.90	14
accuracy			0.89	27
macro avg	0.75	0.67	0.69	27
weighted avg	0.89	0.89	0.88	27

=====

PERFORMANCE ANALYSIS BY: AGE_GROUP

=====

--- Results for Group: 0-17 ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	1.00	0.75	0.86	4
Varicela	0.71	1.00	0.83	5
accuracy			0.83	12
macro avg	0.54	0.55	0.54	12
weighted avg	0.80	0.83	0.80	12

--- Results for Group: 18-59 ---

	precision	recall	f1-score	support
Herpes	1.00	0.75	0.86	4
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.94	0.94	0.94	16
Sarampion	0.95	0.84	0.89	25
Varicela	0.79	0.95	0.86	20

accuracy			0.90	67
macro avg	0.94	0.90	0.91	67
weighted avg	0.91	0.90	0.90	67

--- Results for Group: 60+ ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	0.00	0.00	0.00	0
Varicela	0.00	0.00	0.00	0
accuracy			1.00	1
macro avg	0.20	0.20	0.20	1
weighted avg	1.00	1.00	1.00	1

=====

PERFORMANCE ANALYSIS BY: DOMINANT_RACE

=====

--- Results for Group: asian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	0.00	0.00	0.00	1
Varicela	0.83	1.00	0.91	5
accuracy			0.86	7
macro avg	0.37	0.40	0.38	7
weighted avg	0.74	0.86	0.79	7

--- Results for Group: black ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	1.00	1.00	1.00	1
Varicela	0.00	0.00	0.00	0
accuracy			1.00	4
macro avg	0.40	0.40	0.40	4
weighted avg	1.00	1.00	1.00	4

--- Results for Group: indian ---

	precision	recall	f1-score	support
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Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	0.50	0.67	2
Sarampion	0.86	0.86	0.86	7
Varicela	0.00	0.00	0.00	1
accuracy			0.70	10
macro avg	0.37	0.27	0.30	10
weighted avg	0.80	0.70	0.73	10

--- Results for Group: latino hispanic ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	1.00	1.00	1.00	3
Varicela	0.00	0.00	0.00	0
accuracy			1.00	3
macro avg	0.20	0.20	0.20	3
weighted avg	1.00	1.00	1.00	3

--- Results for Group: middle eastern ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	0.00	0.00	0.00	0
Varicela	0.00	0.00	0.00	0
accuracy			1.00	1
macro avg	0.20	0.20	0.20	1
weighted avg	1.00	1.00	1.00	1

--- Results for Group: white ---

	precision	recall	f1-score	support
Herpes	1.00	0.67	0.80	6
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.92	1.00	0.96	11
Sarampion	1.00	0.82	0.90	17
Varicela	0.83	1.00	0.90	19
accuracy			0.91	55
macro avg	0.95	0.90	0.91	55
weighted avg	0.92	0.91	0.91	55

=====

FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.0000 EOD (max TPR gap): 1.0000

Worst pair: 0-17|female|white vs 18-59|female|indian

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FAIRNESS SUMMARY (FILTERED, n >= 15) | WGA: 0.9310 EOD (max TPR gap): 0.1361
Worst pair (among filtered): 18-59|female|white vs 18-59|male|white
Groups used (filtered): ['18-59|female|white', '18-59|male|white']

=====

#####

####

FOLD 7 ### Train=400, Val=85, Test=80

#####

####

Params: 24.73M | Trainable: 1.22M (4.93%)
Approx MACs: 4.13 G, Params (THOP): 24.73 M

=====

START TRAINING (Multimodal ResNet-50) for 200 epochs

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Epoch 1/200	Train 1.7015/0.2475	Val 1.6104/0.3059	LR 2.00e-05
Epoch 2/200	Train 1.6579/0.2950	Val 1.5956/0.3412	LR 3.00e-05
Epoch 3/200	Train 1.5988/0.3400	Val 1.5687/0.4000	LR 4.00e-05
Epoch 4/200	Train 1.5060/0.4200	Val 1.5262/0.4588	LR 5.00e-05
Epoch 5/200	Train 1.3653/0.4950	Val 1.4648/0.5294	LR 6.00e-05
Epoch 6/200	Train 1.2710/0.5775	Val 1.3833/0.6588	LR 7.00e-05
Epoch 7/200	Train 1.1705/0.6275	Val 1.2853/0.7529	LR 8.00e-05
Epoch 8/200	Train 1.1071/0.6550	Val 1.1776/0.7647	LR 9.00e-05
Epoch 9/200	Train 1.0137/0.7125	Val 1.0706/0.7647	LR 1.00e-04
Epoch 10/200	Train 0.9374/0.7600	Val 0.9806/0.7882	LR 1.00e-04
Epoch 11/200	Train 0.8876/0.7850	Val 0.9120/0.7882	LR 1.00e-04
Epoch 12/200	Train 0.8441/0.8025	Val 0.8630/0.7882	LR 1.00e-04
Epoch 13/200	Train 0.7996/0.8400	Val 0.8284/0.7765	LR 9.99e-05
Epoch 14/200	Train 0.7746/0.8475	Val 0.8035/0.7765	LR 9.99e-05
Epoch 15/200	Train 0.7571/0.8525	Val 0.7879/0.7882	LR 9.98e-05
Epoch 16/200	Train 0.7388/0.8775	Val 0.7724/0.8000	LR 9.98e-05
Epoch 17/200	Train 0.7147/0.8775	Val 0.7673/0.8000	LR 9.97e-05
Epoch 18/200	Train 0.6890/0.8775	Val 0.7626/0.8118	LR 9.96e-05
Epoch 19/200	Train 0.6840/0.9050	Val 0.7566/0.8118	LR 9.94e-05
Epoch 20/200	Train 0.6580/0.8950	Val 0.7558/0.8000	LR 9.93e-05
Epoch 21/200	Train 0.6573/0.9025	Val 0.7553/0.8118	LR 9.92e-05
Epoch 22/200	Train 0.6403/0.9075	Val 0.7551/0.8118	LR 9.90e-05
Epoch 23/200	Train 0.6467/0.9125	Val 0.7547/0.8118	LR 9.88e-05
Epoch 24/200	Train 0.6302/0.9100	Val 0.7522/0.8000	LR 9.87e-05
Epoch 25/200	Train 0.5965/0.9400	Val 0.7488/0.8000	LR 9.85e-05
Epoch 26/200	Train 0.6017/0.9250	Val 0.7493/0.8000	LR 9.83e-05
Epoch 27/200	Train 0.6034/0.9250	Val 0.7481/0.8000	LR 9.80e-05
Epoch 28/200	Train 0.5812/0.9425	Val 0.7458/0.8000	LR 9.78e-05
Epoch 29/200	Train 0.5901/0.9375	Val 0.7461/0.8000	LR 9.76e-05
Epoch 30/200	Train 0.5731/0.9450	Val 0.7457/0.8000	LR 9.73e-05
Epoch 31/200	Train 0.5774/0.9350	Val 0.7444/0.8000	LR 9.70e-05
Epoch 32/200	Train 0.5639/0.9500	Val 0.7444/0.8000	LR 9.67e-05
Epoch 33/200	Train 0.5695/0.9325	Val 0.7465/0.8000	LR 9.64e-05
Epoch 34/200	Train 0.5649/0.9475	Val 0.7452/0.8000	LR 9.61e-05
Epoch 35/200	Train 0.5662/0.9375	Val 0.7473/0.8000	LR 9.58e-05
Epoch 36/200	Train 0.5421/0.9650	Val 0.7471/0.8000	LR 9.55e-05

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Epoch 37/200 | Train 0.5768/0.9375 | Val 0.7439/0.8000 | LR 9.51e-05
Epoch 38/200 | Train 0.5375/0.9600 | Val 0.7419/0.8000 | LR 9.47e-05
Epoch 39/200 | Train 0.5387/0.9575 | Val 0.7380/0.8000 | LR 9.44e-05
Epoch 40/200 | Train 0.5324/0.9550 | Val 0.7346/0.8000 | LR 9.40e-05
Epoch 41/200 | Train 0.5300/0.9550 | Val 0.7345/0.8000 | LR 9.36e-05
Epoch 42/200 | Train 0.5290/0.9500 | Val 0.7342/0.8000 | LR 9.32e-05
Epoch 43/200 | Train 0.5349/0.9700 | Val 0.7355/0.8000 | LR 9.27e-05
Epoch 44/200 | Train 0.5224/0.9600 | Val 0.7353/0.8000 | LR 9.23e-05
Epoch 45/200 | Train 0.5299/0.9675 | Val 0.7310/0.8000 | LR 9.19e-05
Epoch 46/200 | Train 0.5258/0.9650 | Val 0.7329/0.8000 | LR 9.14e-05
Epoch 47/200 | Train 0.5134/0.9550 | Val 0.7312/0.8118 | LR 9.09e-05
Epoch 48/200 | Train 0.5036/0.9750 | Val 0.7312/0.8118 | LR 9.05e-05
Epoch 49/200 | Train 0.5226/0.9675 | Val 0.7315/0.8118 | LR 9.00e-05
Epoch 50/200 | Train 0.5094/0.9675 | Val 0.7311/0.8118 | LR 8.95e-05
Epoch 51/200 | Train 0.5244/0.9550 | Val 0.7301/0.8118 | LR 8.89e-05
Epoch 52/200 | Train 0.5087/0.9725 | Val 0.7295/0.8118 | LR 8.84e-05
Epoch 53/200 | Train 0.5157/0.9725 | Val 0.7274/0.8235 | LR 8.79e-05
Epoch 54/200 | Train 0.5044/0.9750 | Val 0.7302/0.8235 | LR 8.73e-05
Epoch 55/200 | Train 0.5022/0.9800 | Val 0.7329/0.8235 | LR 8.68e-05
Epoch 56/200 | Train 0.4971/0.9775 | Val 0.7329/0.8353 | LR 8.62e-05
Epoch 57/200 | Train 0.5032/0.9750 | Val 0.7316/0.8353 | LR 8.56e-05
Epoch 58/200 | Train 0.4996/0.9725 | Val 0.7366/0.8235 | LR 8.51e-05
Epoch 59/200 | Train 0.4998/0.9675 | Val 0.7354/0.8235 | LR 8.45e-05
Epoch 60/200 | Train 0.4919/0.9700 | Val 0.7335/0.8235 | LR 8.39e-05
Epoch 61/200 | Train 0.4866/0.9750 | Val 0.7363/0.8235 | LR 8.33e-05
Epoch 62/200 | Train 0.4960/0.9800 | Val 0.7394/0.8235 | LR 8.26e-05
Epoch 63/200 | Train 0.4910/0.9775 | Val 0.7409/0.8353 | LR 8.20e-05
Epoch 64/200 | Train 0.4903/0.9750 | Val 0.7421/0.8235 | LR 8.14e-05
Epoch 65/200 | Train 0.4806/0.9850 | Val 0.7456/0.8353 | LR 8.07e-05
Epoch 66/200 | Train 0.5061/0.9725 | Val 0.7459/0.8235 | LR 8.01e-05
Epoch 67/200 | Train 0.4918/0.9775 | Val 0.7464/0.8353 | LR 7.94e-05
Epoch 68/200 | Train 0.4781/0.9850 | Val 0.7434/0.8235 | LR 7.87e-05
Epoch 69/200 | Train 0.4966/0.9650 | Val 0.7390/0.8235 | LR 7.80e-05
Epoch 70/200 | Train 0.4742/0.9875 | Val 0.7391/0.8235 | LR 7.73e-05
Epoch 71/200 | Train 0.4816/0.9775 | Val 0.7341/0.8118 | LR 7.67e-05
Epoch 72/200 | Train 0.4831/0.9850 | Val 0.7336/0.8118 | LR 7.59e-05
Epoch 73/200 | Train 0.4722/0.9900 | Val 0.7324/0.8235 | LR 7.52e-05

```

Early stopping.

Training complete in 4.5 minutes. Best Val Loss 0.7274

=====

OVERALL PERFORMANCE (ON COMBINED DATASET)

=====

	precision	recall	f1-score	support
Herpes	0.60	0.50	0.55	6
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.85	0.94	0.89	18
Sarampion	0.92	0.83	0.87	29
Varicela	0.89	0.96	0.92	25
accuracy			0.88	80
macro avg	0.85	0.85	0.85	80
weighted avg	0.87	0.88	0.87	80

```
=====
PERFORMANCE ANALYSIS BY: DOMINANT_GENDER
=====
```

```
--- Results for Group: Man ---
```

	precision	recall	f1-score	support
Herpes	0.60	0.50	0.55	6
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.82	0.93	0.88	15
Sarampion	0.83	0.71	0.77	14
Varicela	0.86	0.92	0.89	13
accuracy			0.82	50
macro avg	0.82	0.81	0.82	50
weighted avg	0.82	0.82	0.81	50

```
--- Results for Group: Woman ---
```

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	1.00	0.93	0.97	15
Varicela	0.92	1.00	0.96	12
accuracy			0.97	30
macro avg	0.58	0.59	0.59	30
weighted avg	0.97	0.97	0.97	30

```
=====
PERFORMANCE ANALYSIS BY: AGE_GROUP
=====
```

```
--- Results for Group: 0-17 ---
```

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.67	1.00	0.80	2
Sarampion	1.00	0.67	0.80	3
Varicela	0.83	1.00	0.91	5
accuracy			0.82	11
macro avg	0.50	0.53	0.50	11
weighted avg	0.77	0.82	0.78	11

```
--- Results for Group: 18-59 ---
```

	precision	recall	f1-score	support
Herpes	0.60	0.60	0.60	5

Melanoma	1.00	1.00	1.00	2
Monkeypox	0.88	0.93	0.90	15
Sarampion	0.92	0.85	0.88	26
Varicela	0.90	0.95	0.93	20
accuracy			0.88	68
macro avg	0.86	0.87	0.86	68
weighted avg	0.88	0.88	0.88	68

--- Results for Group: 60+ ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	0.00	0.00	0.00	0
Varicela	0.00	0.00	0.00	0
accuracy			1.00	1
macro avg	0.20	0.20	0.20	1
weighted avg	1.00	1.00	1.00	1

=====

PERFORMANCE ANALYSIS BY: DOMINANT_RACE

=====

--- Results for Group: asian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	4
Sarampion	1.00	1.00	1.00	5
Varicela	1.00	1.00	1.00	4
accuracy			1.00	13
macro avg	0.60	0.60	0.60	13
weighted avg	1.00	1.00	1.00	13

--- Results for Group: black ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.71	1.00	0.83	5
Sarampion	0.50	0.50	0.50	2
Varicela	0.00	0.00	0.00	1
accuracy			0.67	9
macro avg	0.24	0.30	0.27	9
weighted avg	0.51	0.67	0.57	9

```

--- Results for Group: indian ---
      precision    recall  f1-score   support

   Herpes         0.00      0.00      0.00         0
   Melanoma        0.00      0.00      0.00         0
 Monkeypox        1.00      1.00      1.00         4
 Sarampion        1.00      1.00      1.00         1
   Varicela        0.00      0.00      0.00         0

 accuracy                   1.00         5
 macro avg         0.40      0.40      0.40         5
 weighted avg      1.00      1.00      1.00         5

```

```

--- Results for Group: latino hispanic ---
      precision    recall  f1-score   support

   Herpes         0.00      0.00      0.00         0
   Melanoma        0.00      0.00      0.00         0
 Monkeypox        1.00      1.00      1.00         2
 Sarampion        1.00      1.00      1.00         3
   Varicela        0.00      0.00      0.00         0

 accuracy                   1.00         5
 macro avg         0.40      0.40      0.40         5
 weighted avg      1.00      1.00      1.00         5

```

```

--- Results for Group: middle eastern ---
      precision    recall  f1-score   support

   Herpes         0.00      0.00      0.00         0
   Melanoma        0.00      0.00      0.00         0
 Monkeypox        0.00      0.00      0.00         0
 Sarampion        1.00      0.50      0.67         2
   Varicela        0.00      0.00      0.00         0

 accuracy                   0.50         2
 macro avg         0.20      0.10      0.13         2
 weighted avg      1.00      0.50      0.67         2

```

```

--- Results for Group: white ---
      precision    recall  f1-score   support

   Herpes         0.60      0.60      0.60         5
   Melanoma        1.00      1.00      1.00         2
 Monkeypox        1.00      0.67      0.80         3
 Sarampion        0.93      0.81      0.87        16
   Varicela        0.87      1.00      0.93        20

 accuracy                   0.87        46
 macro avg         0.88      0.82      0.84        46
 weighted avg      0.87      0.87      0.87        46

```

```
=====
FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.5000 EOD (max TPR gap): 1.0000
Worst pair: 0-17|female|white vs 0-17|male|black
=====
```

```
=====
FAIRNESS SUMMARY (FILTERED, n >= 15) | WGA: 0.7778 EOD (max TPR gap): 0.4583
Worst pair (among filtered): 18-59|female|white vs 18-59|male|white
Groups used (filtered): ['18-59|female|white', '18-59|male|white']
=====
```

```
=====
OVERALL K-FOLD RESULTS
=====
```

```
Accuracy      : 0.8673 ± 0.0393
ROC AUC (Macro) : 0.9761 ± 0.0099
Cohen's Kappa  : 0.8140 ± 0.0553
MCC           : 0.8165 ± 0.0549
-----
```

```
WGA (All Groups) : 0.1429 ± 0.2259
EOD (All Groups) : 1.0000 ± 0.0000
WGA (n>=15) : 0.8267 ± 0.1046
EOD (n>=15) : 0.2159 ± 0.1499
```

```
=====
POOLED SUBGROUP ACCURACY ACROSS ALL FOLDS (SUPPORT >= 15)
=====
```

	Group	Accuracy	Support
0	age:0-17	0.786885	61
1	age:18-59	0.876248	501
2	composite:0-17 female white	0.866667	15
3	composite:0-17 male white	0.800000	30
4	composite:18-59 female asian	0.884615	26
5	composite:18-59 female latino hispanic	0.920000	25
6	composite:18-59 female white	0.928571	126
7	composite:18-59 male asian	0.820513	39
8	composite:18-59 male black	0.900000	30
9	composite:18-59 male indian	0.941176	34
10	composite:18-59 male latino hispanic	1.000000	23
11	composite:18-59 male middle eastern	0.777778	18
12	composite:18-59 male white	0.831325	166
13	gender:female	0.895238	210
14	gender:male	0.850704	355
15	skin_tone:asian	0.842857	70
16	skin_tone:black	0.868421	38
17	skin_tone:indian	0.840000	50
18	skin_tone:latino hispanic	0.960000	50
19	skin_tone:middle eastern	0.789474	19
20	skin_tone:white	0.866864	338

```
Results saved to 'results_Multimodal_ResNet50_KFold'
```

In []:

In []: